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Tsai et al.

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(54) **OPTICAL IMAGING LENS ASSEMBLY**

(75) Inventors: **Tsung-Han Tsai**, Taichung (TW);
Ming-Ta Chou, Taichung (TW)

(73) Assignee: **Largan Precision Co.**, Taichung (TW)

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G02B 13/18 (2006.01)

G02B 9/60 (2006.01)

(52) **U.S. Cl.** **359/714; 359/764**

(58) **Field of Classification Search** 359/763,
359/764, 714

See application file for complete search history.

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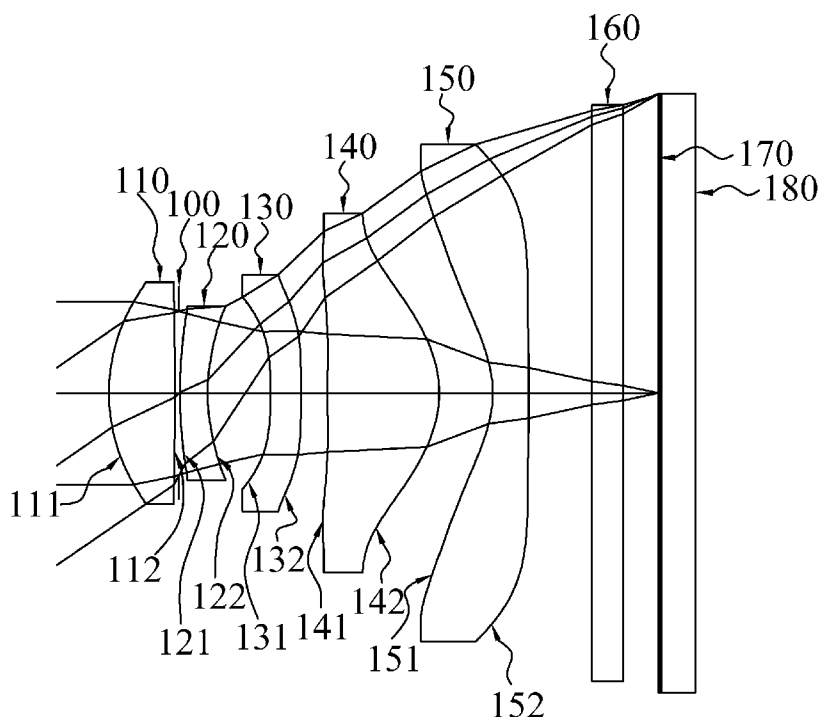
Primary Examiner — Scott J Sugarman

(74) *Attorney, Agent, or Firm* — Wang Law Firm, Inc.; Li K. Wang; Stephen Hsu

(57) **ABSTRACT**

An optical imaging lens assembly comprising five non-cemented lens elements with refractive power, sequentially arranged from an object side to an image side, comprising: the first lens element with positive refractive power, the second lens element with negative refractive power, the third lens element with refractive power having a concave object-side surface and at least one aspheric surface, the fourth lens element with positive refractive power having two aspheric surfaces, and the fifth lens element with negative refractive power having a concave object-side surface and a convex image-side surface with both being aspheric. Wherein an image sensor disposed on an image plane and a stop are also provided. By such arrangements, the image pickup optical system satisfies conditions related to shorten the total length and to reduce the sensitivity for use in compact cameras and mobile phones with camera functionalities.

25 Claims, 20 Drawing Sheets



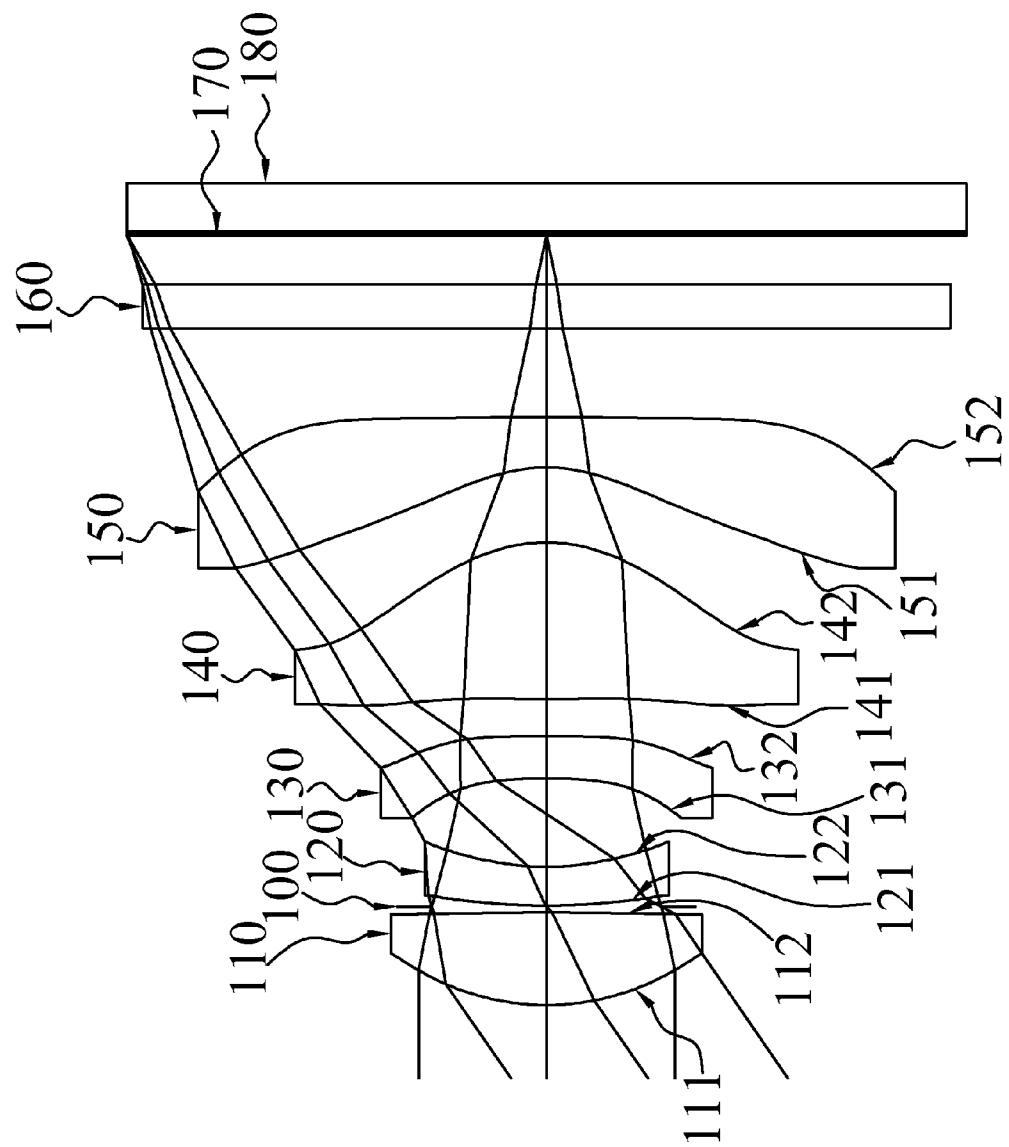


FIG. 1A

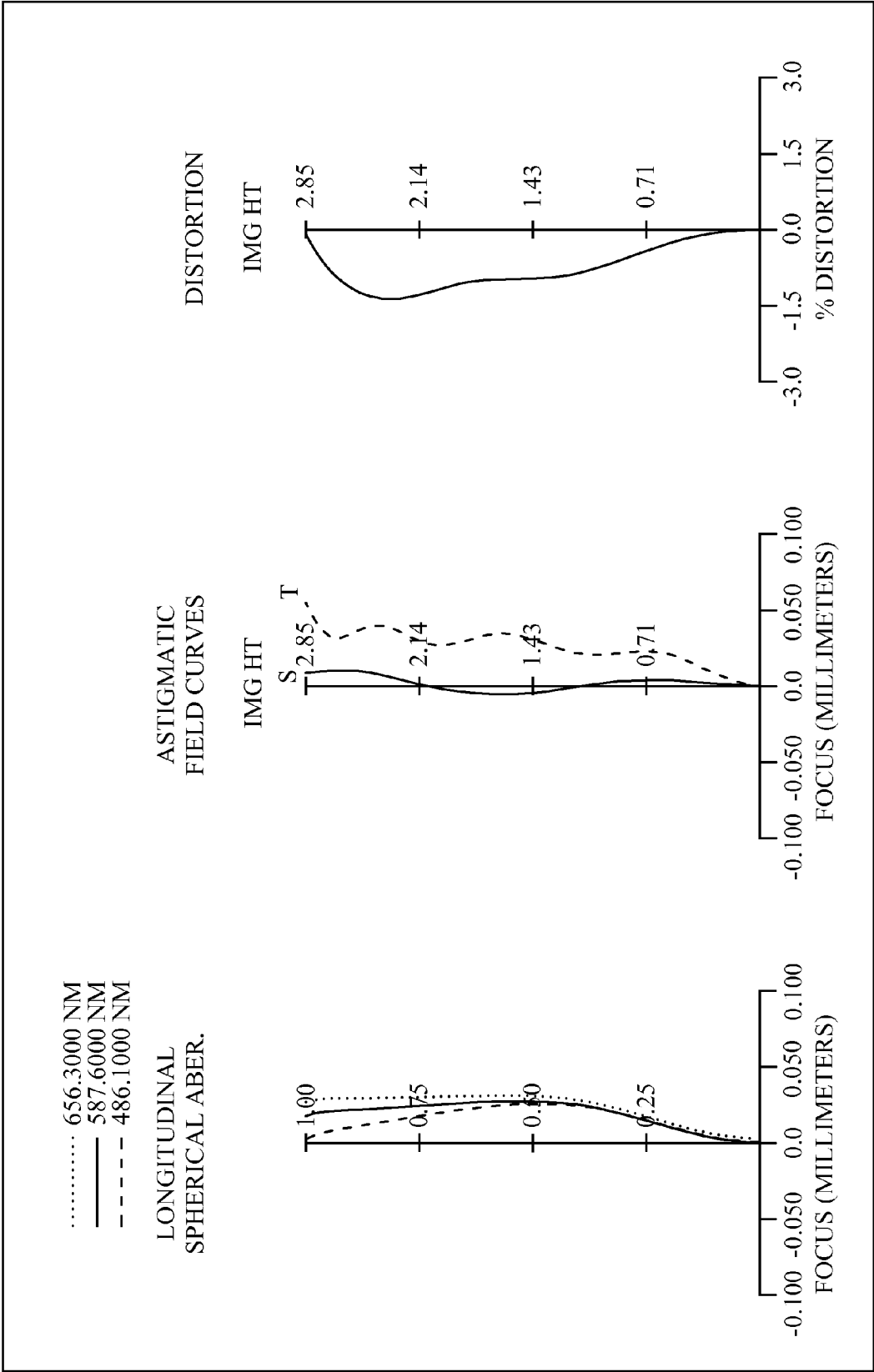


FIG. 1B

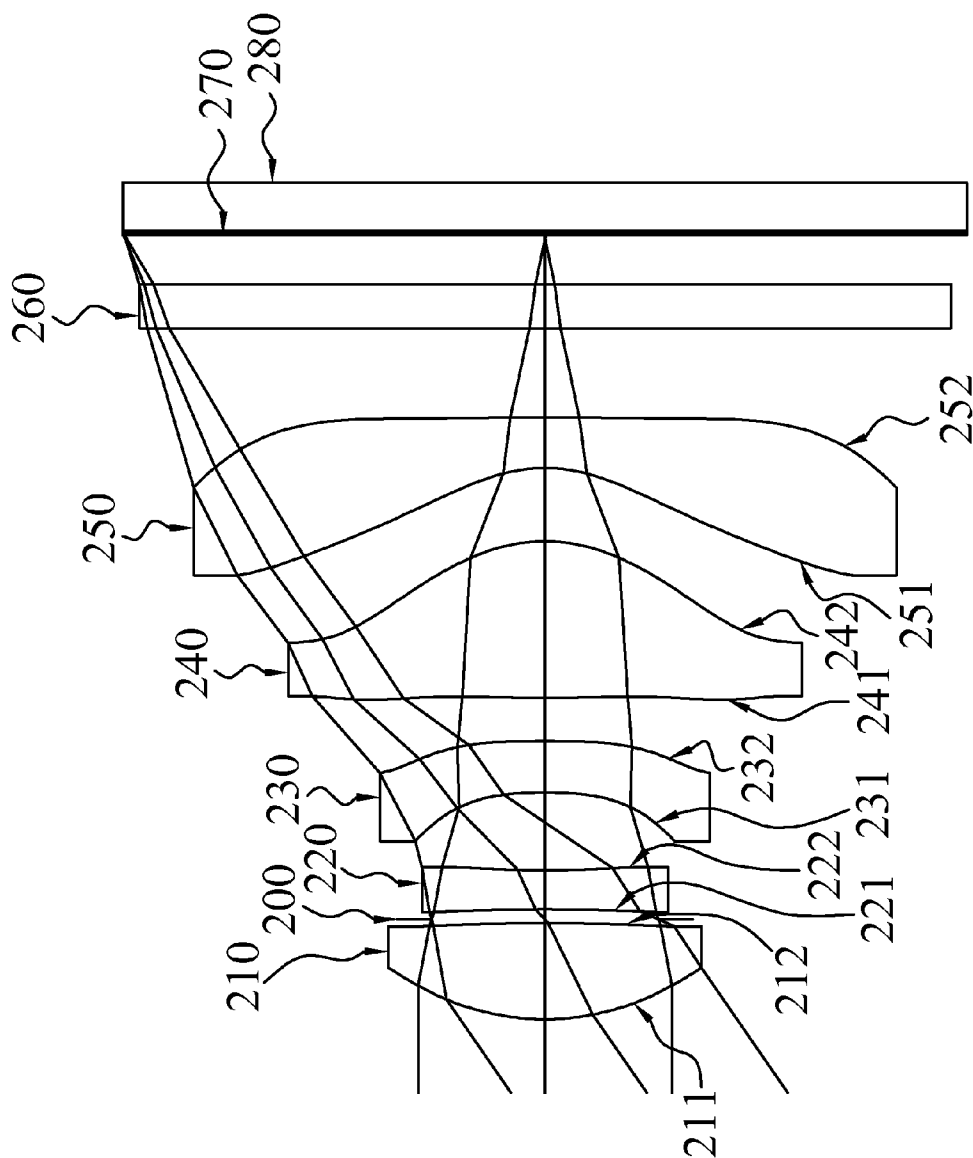


FIG. 2A

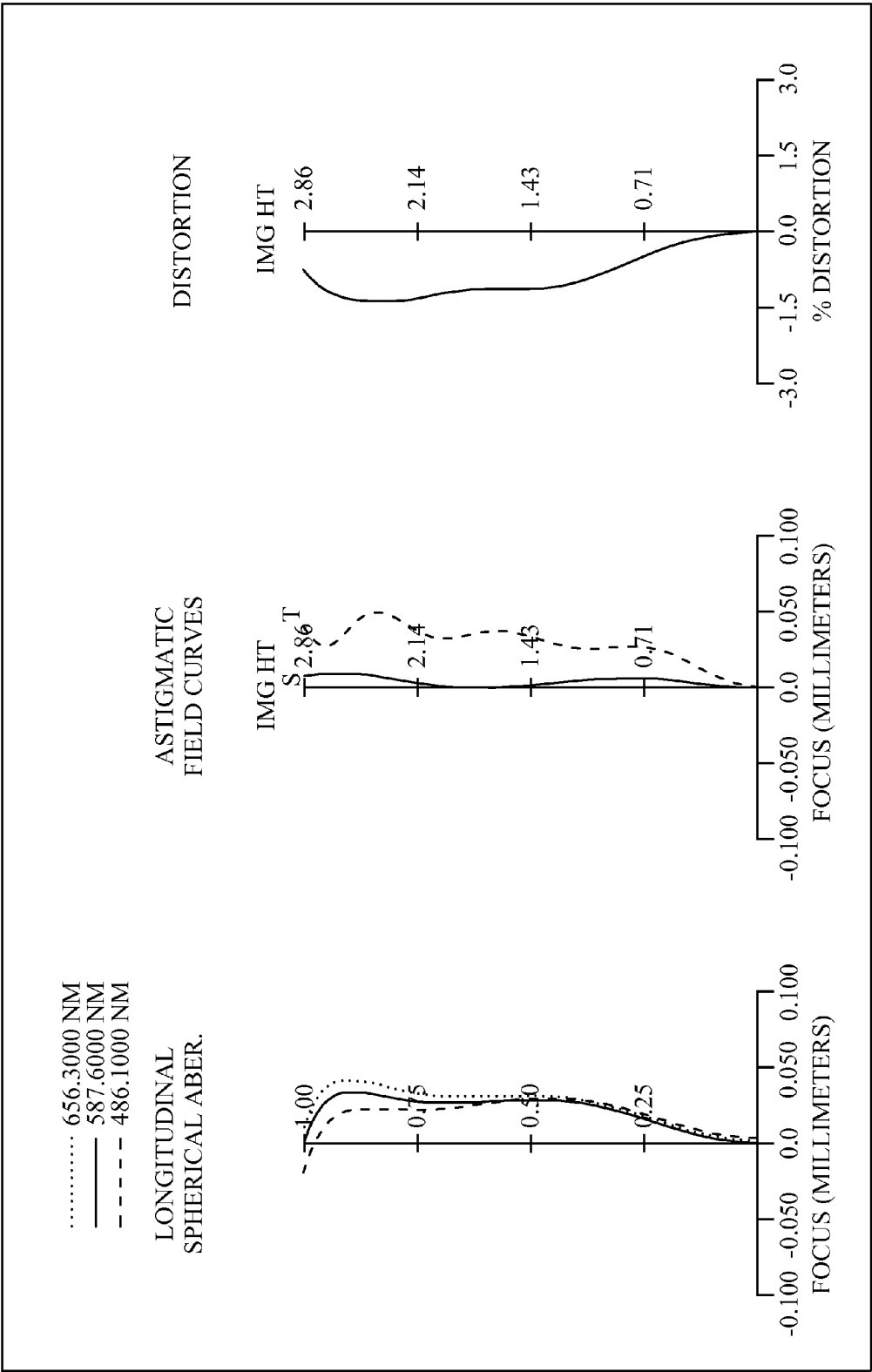


FIG. 2B

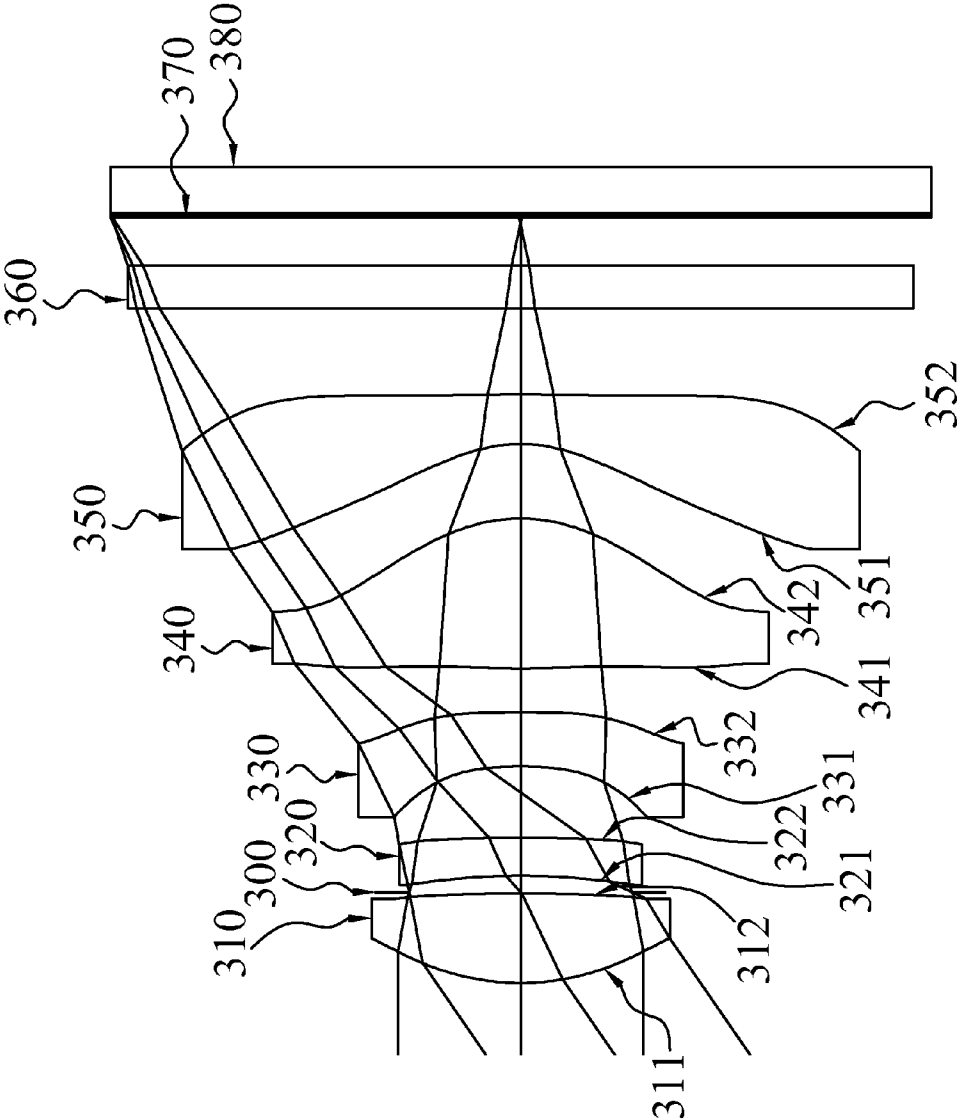
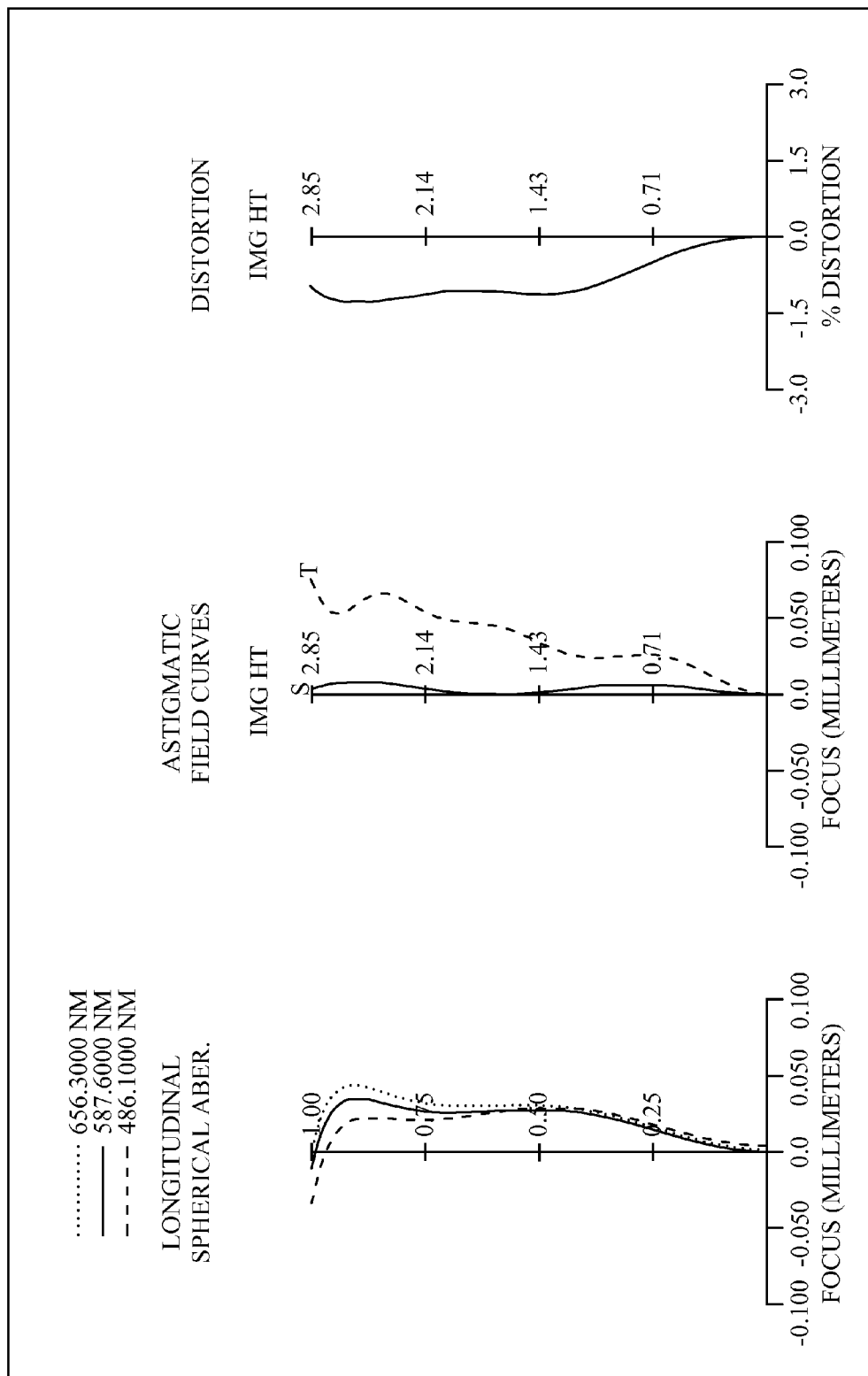


FIG. 3A



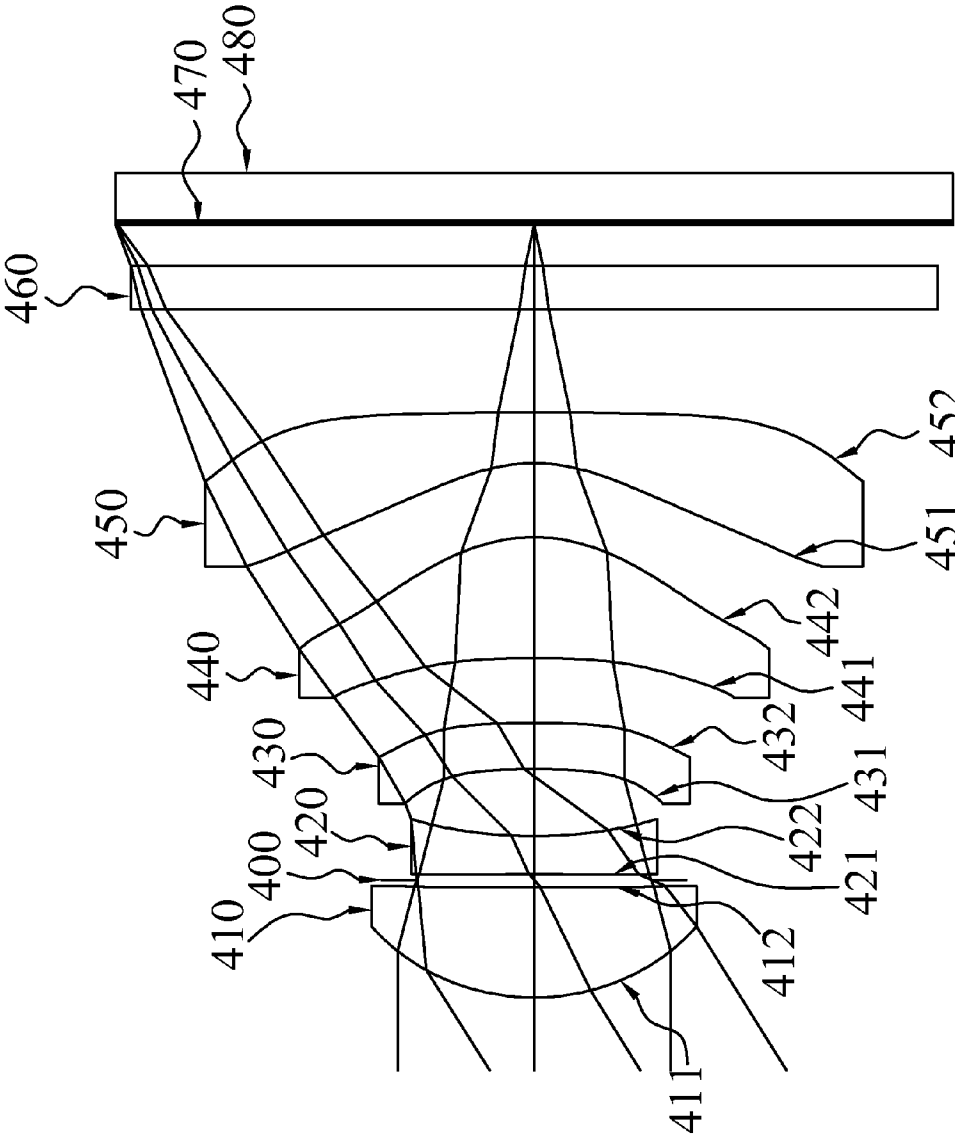


FIG. 4A

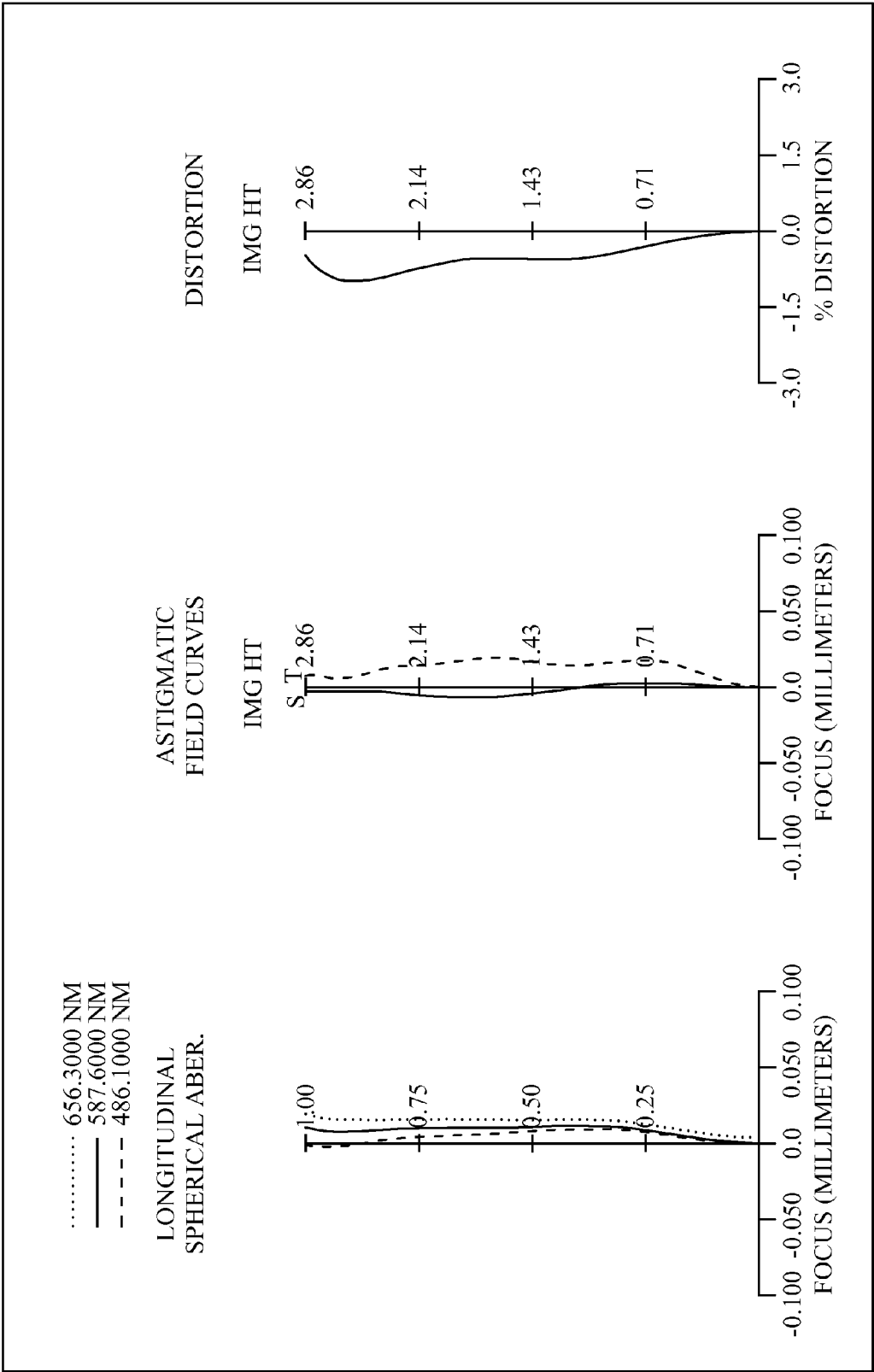


FIG. 4B

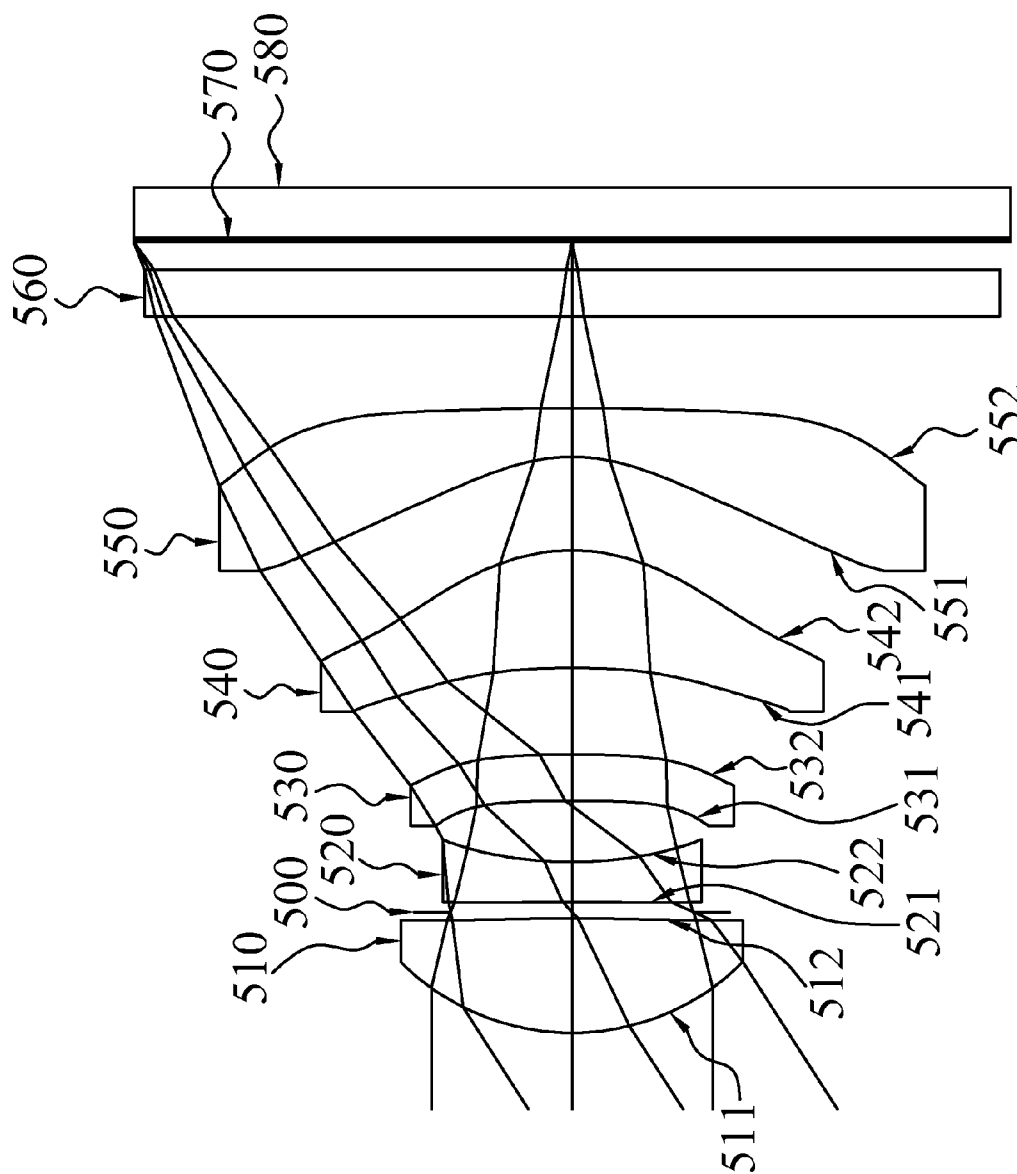


FIG. 5A

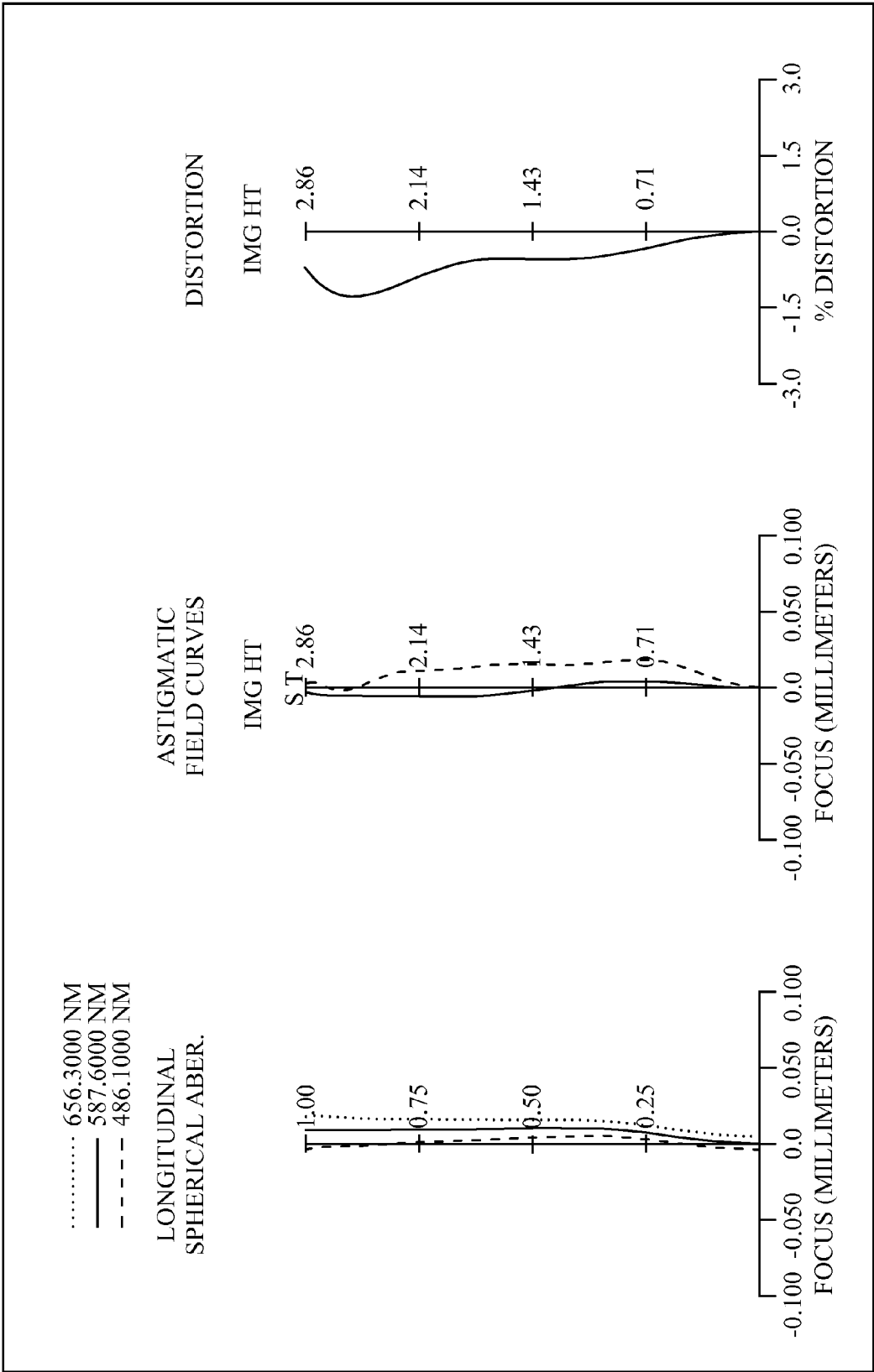


FIG. 5B

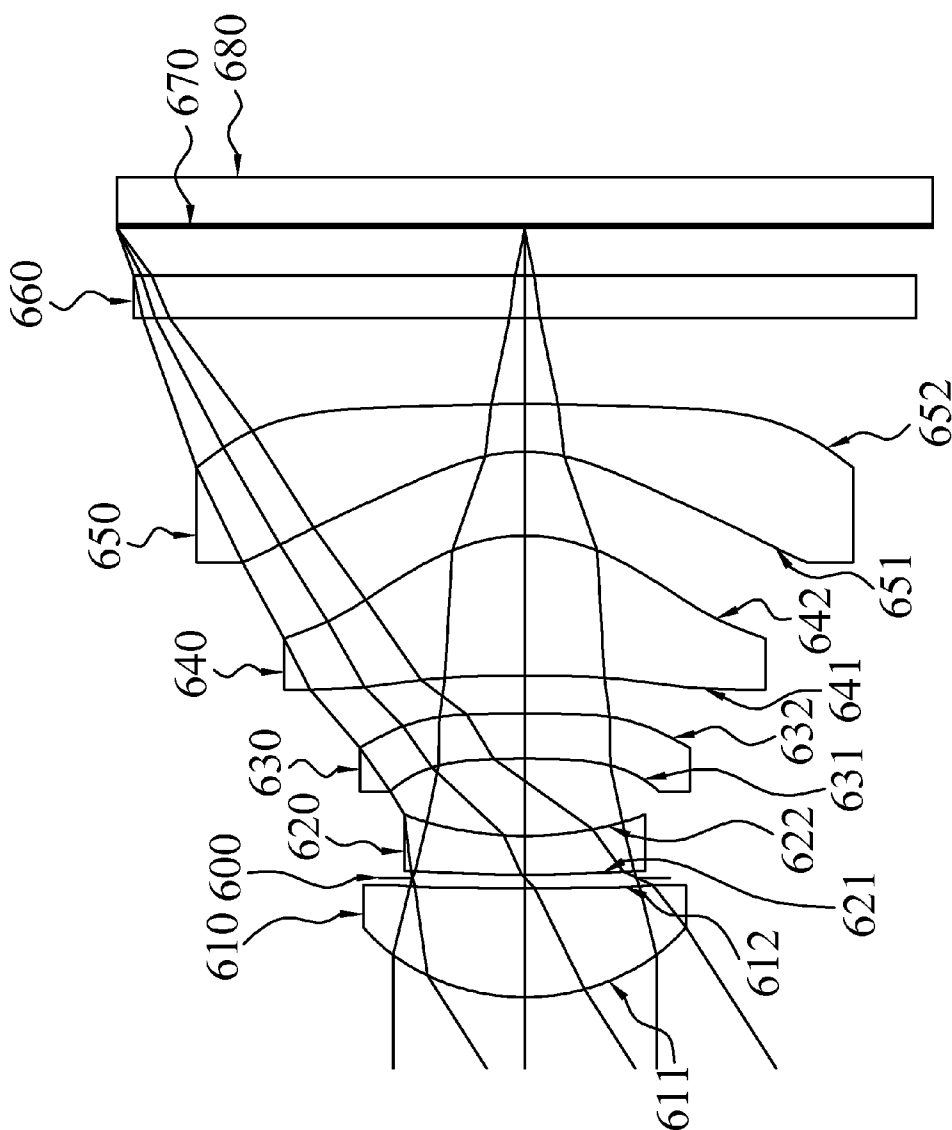


FIG. 6A

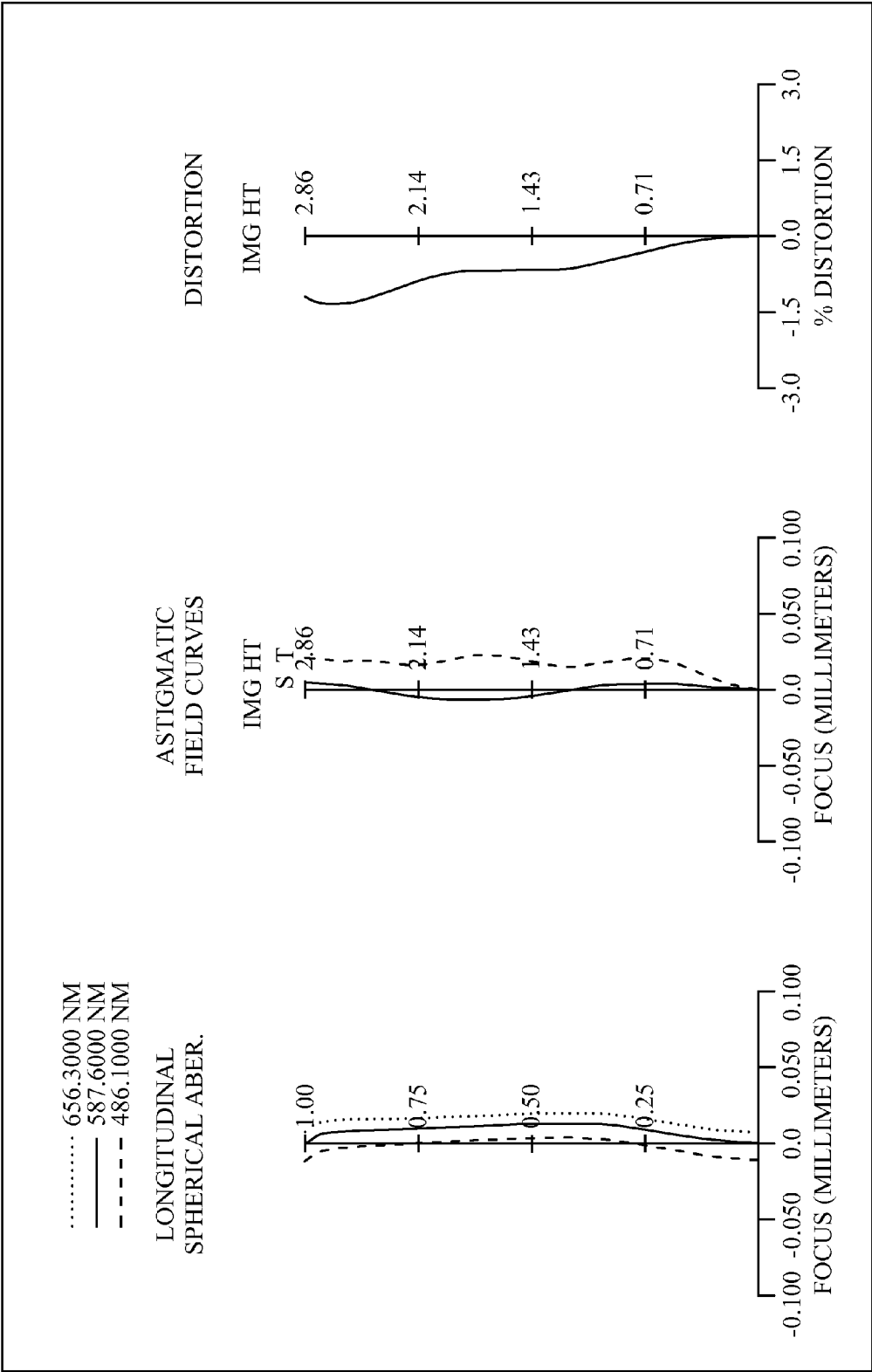


FIG. 6B

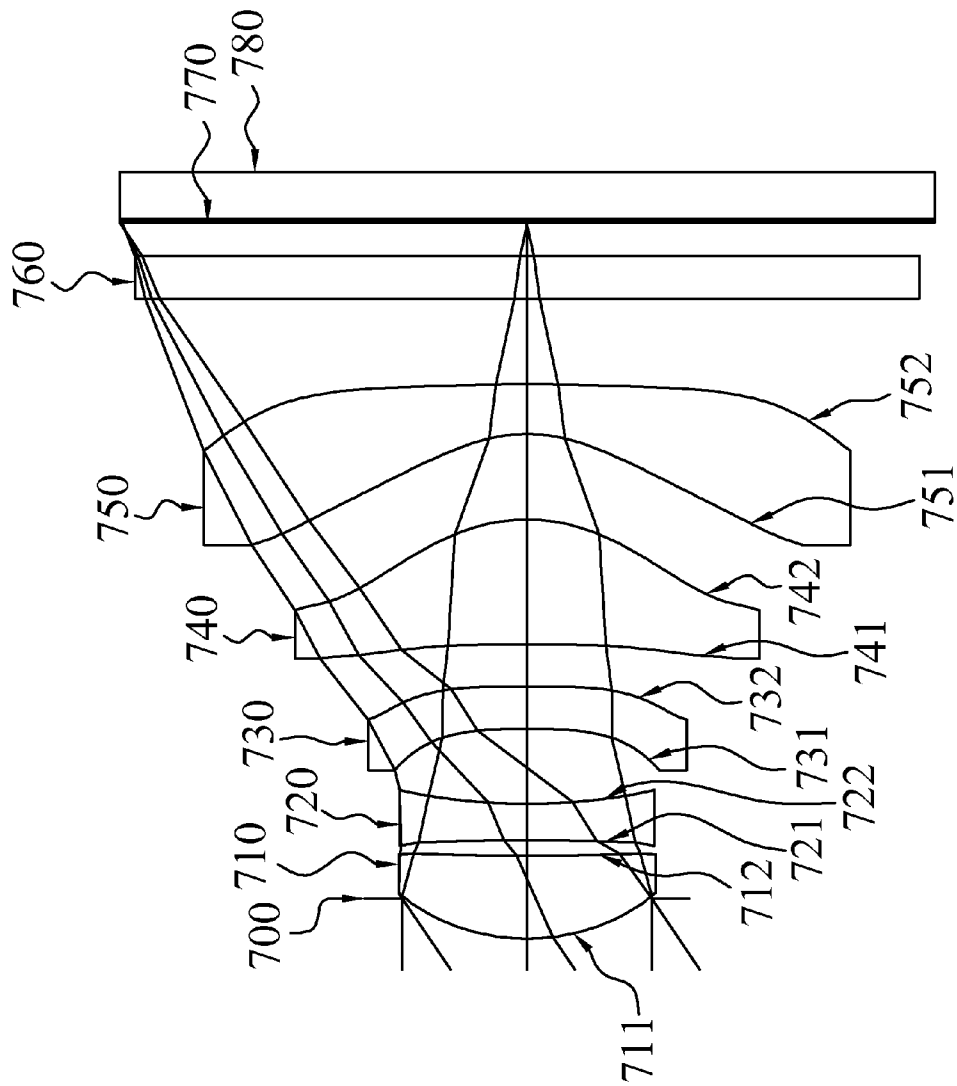


FIG. 7A

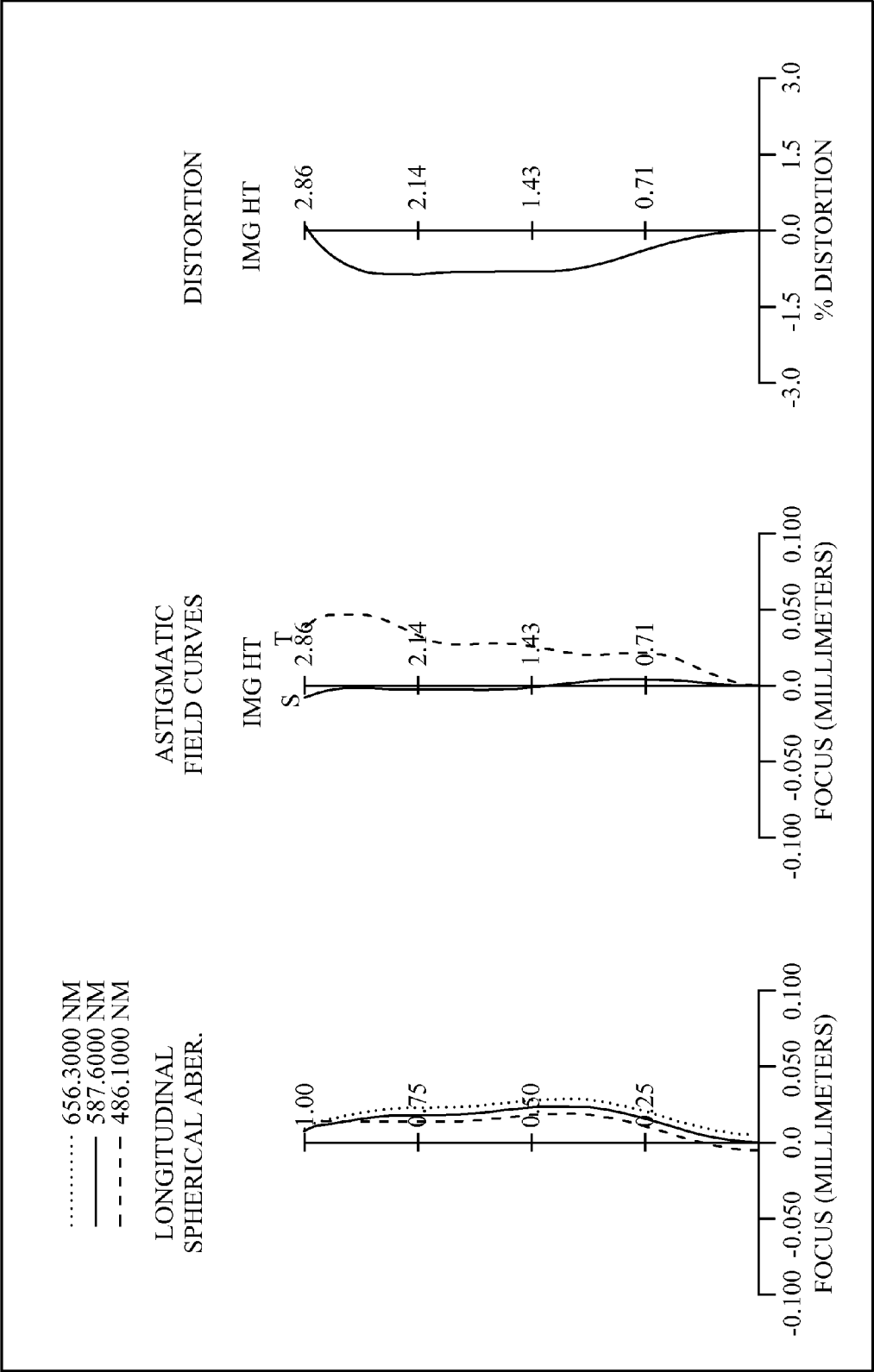


FIG. 7B

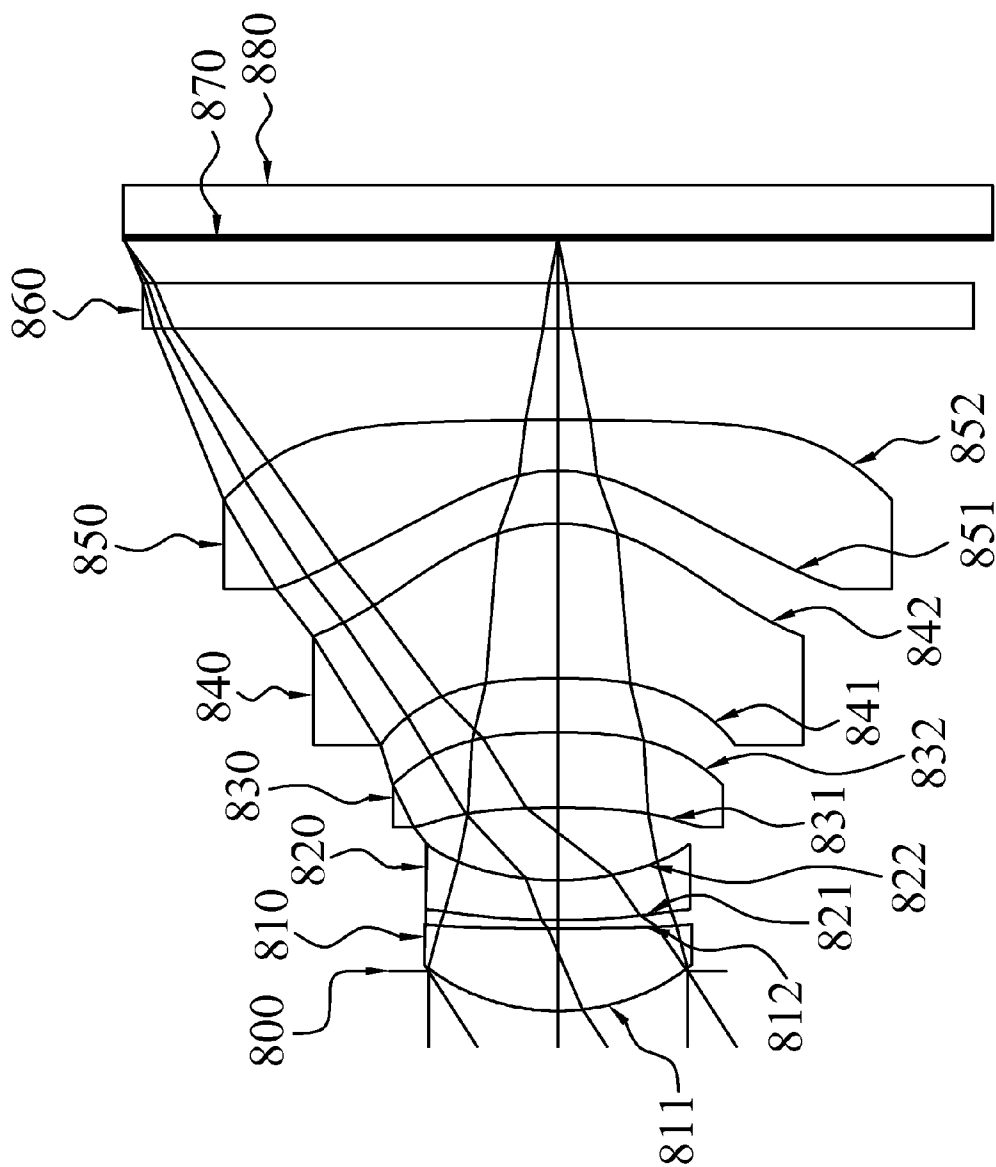


FIG. 8A

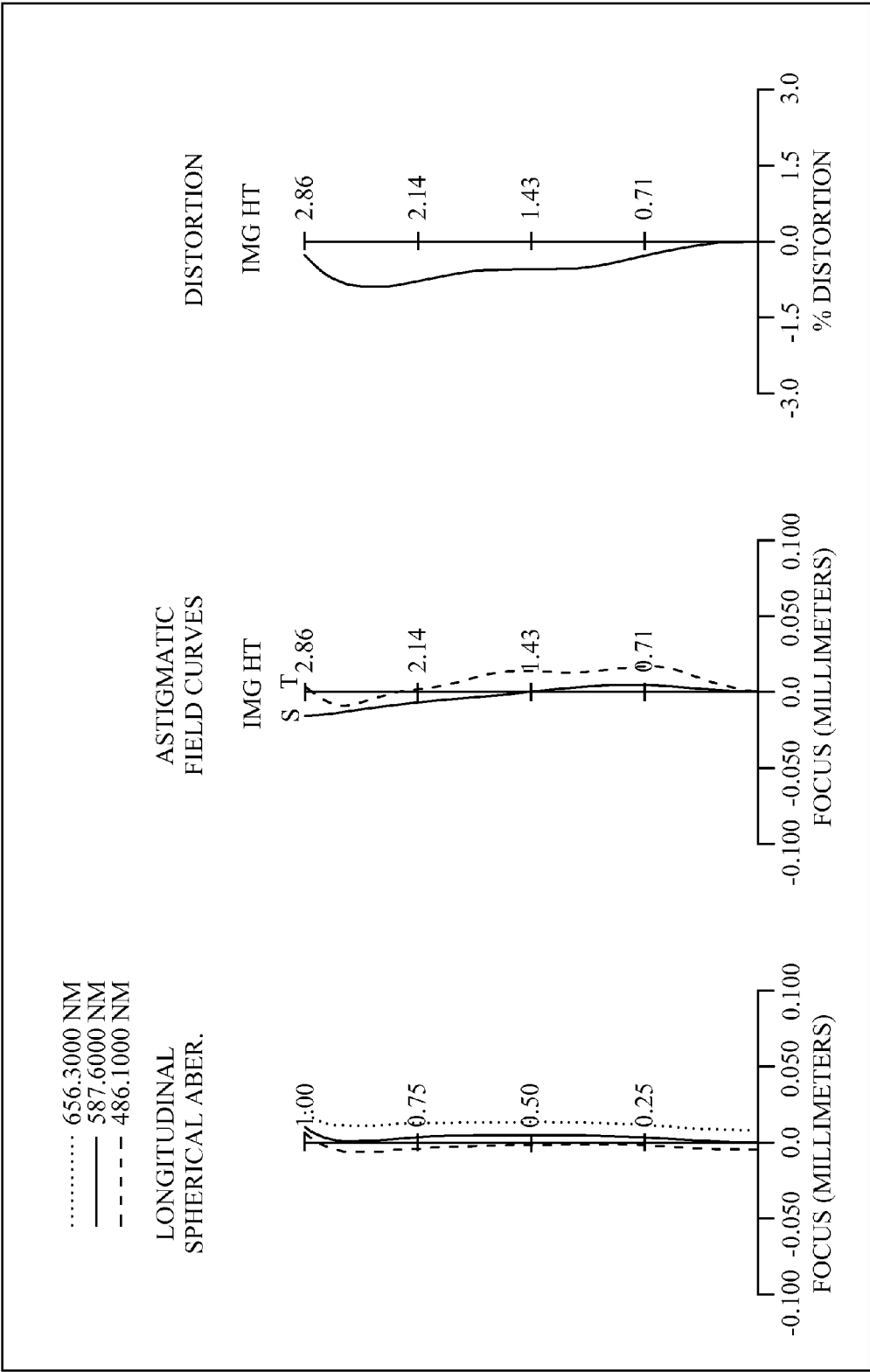


FIG. 8B

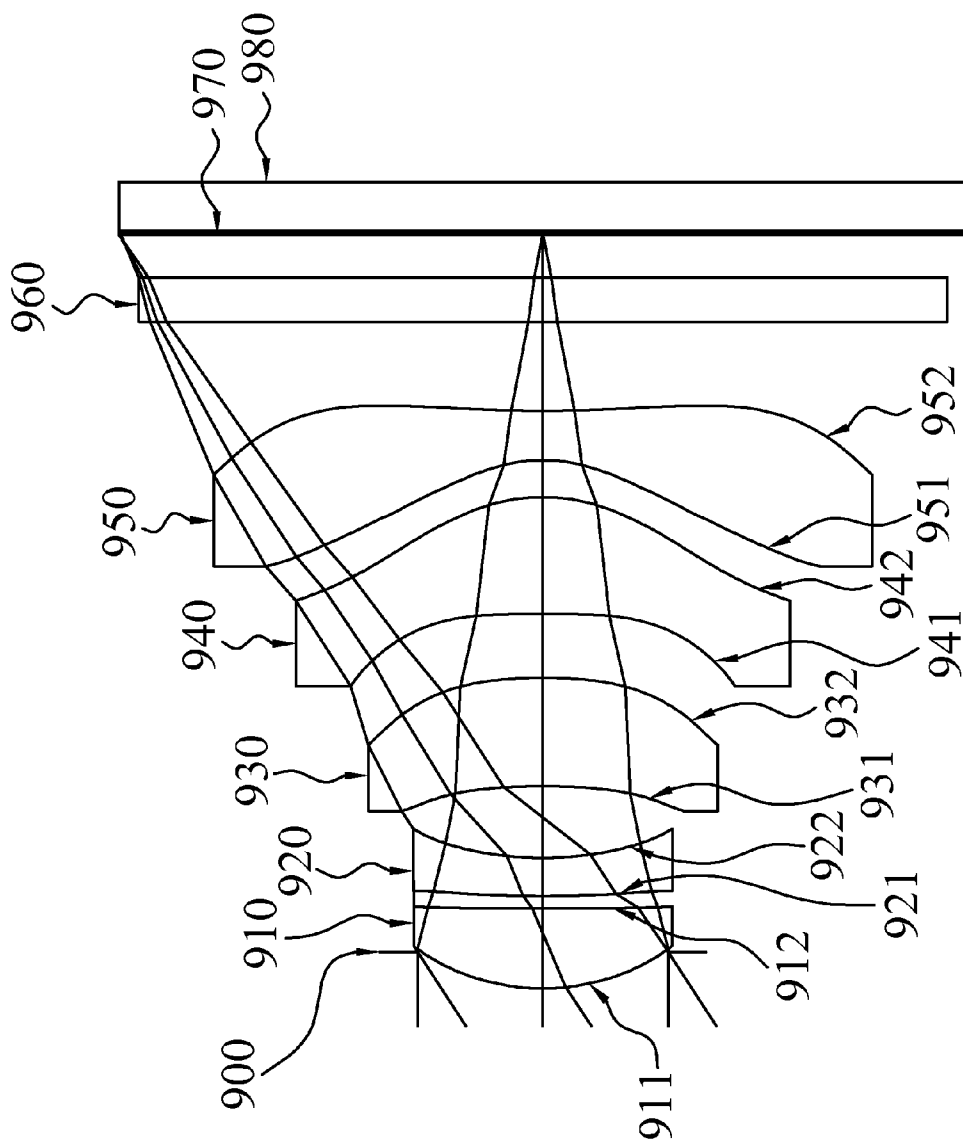


FIG. 9A

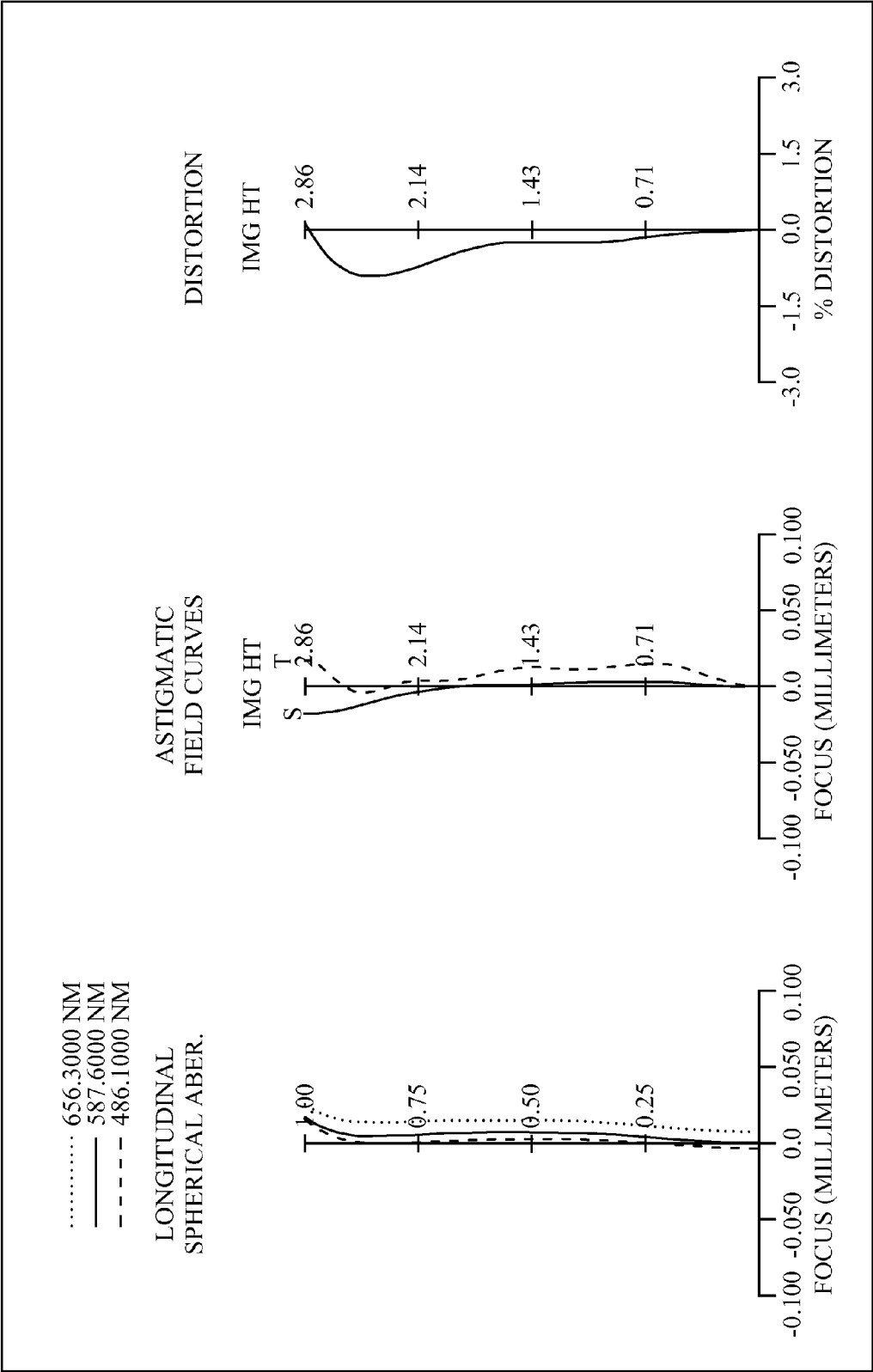


FIG. 9B

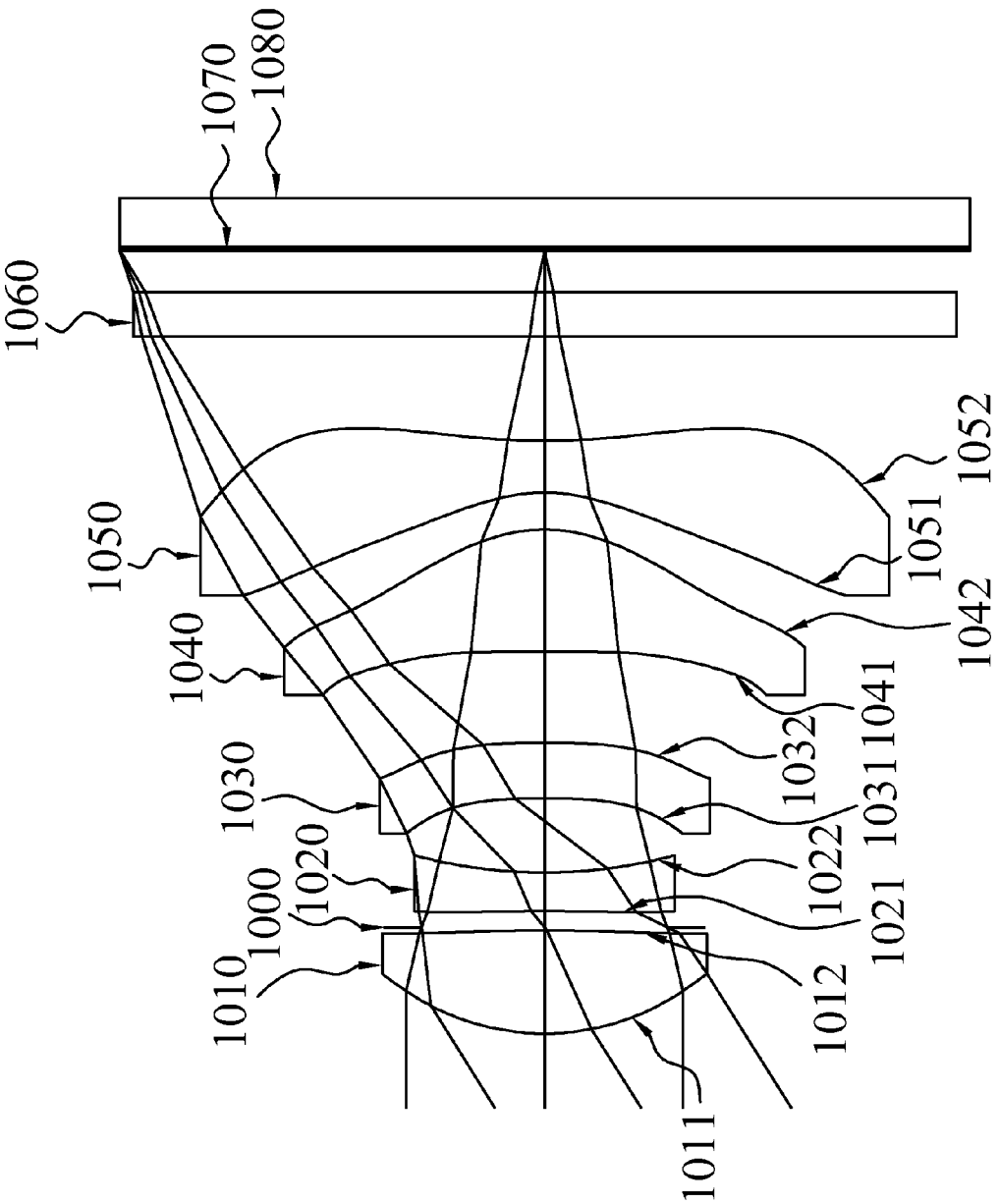


FIG. 10A

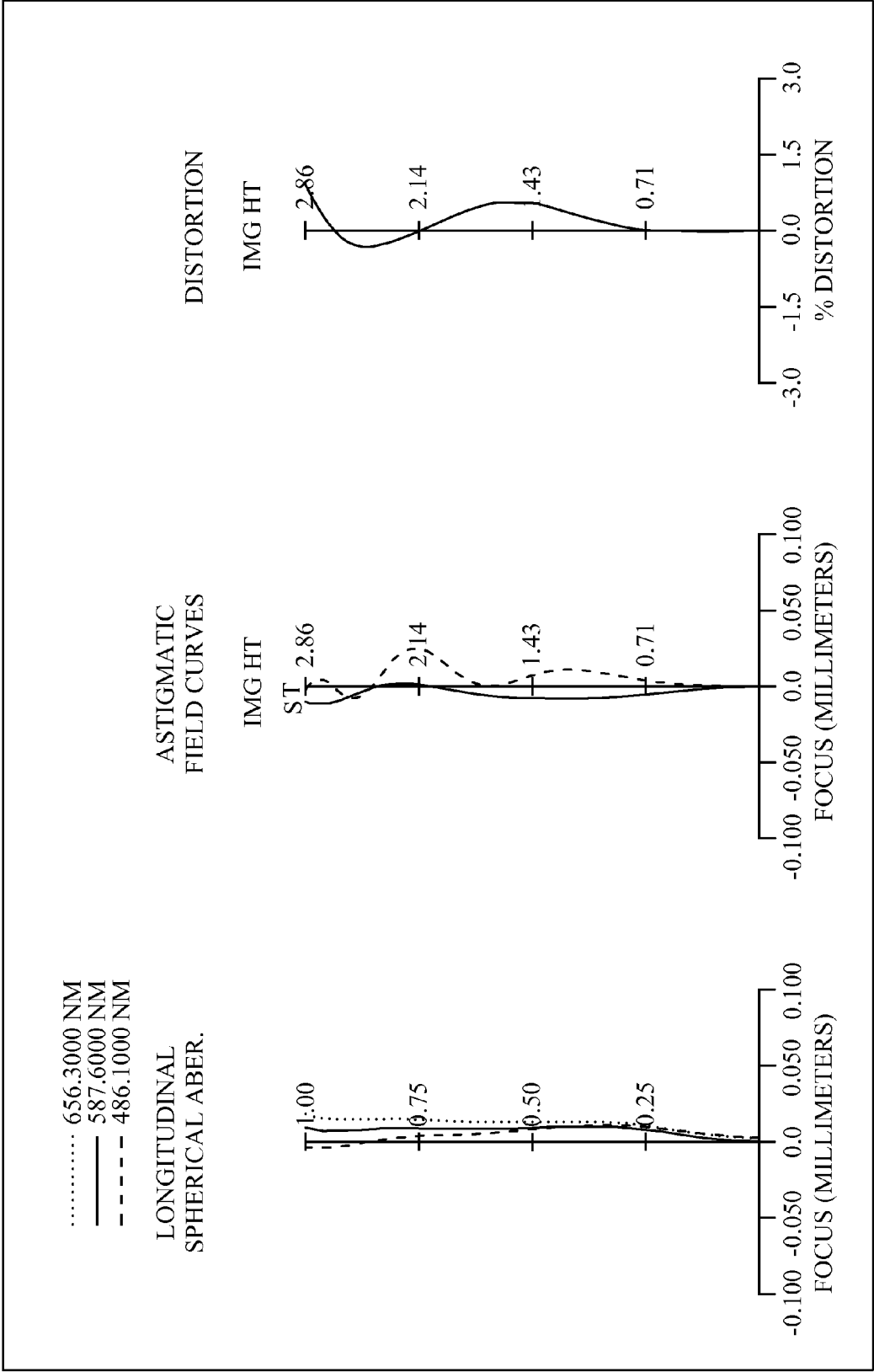


FIG. 10B

OPTICAL IMAGING LENS ASSEMBLY

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to an optical imaging lens assembly, and more particularly to the optical imaging lens assembly comprised of five non-cemented lenses with refractive power to provide a short total length with great image quality for electronic products.

2. Description of the Related Art

In compact electronic products such as digital still cameras and mobile phone cameras, an optical imaging lens assembly is generally installed for capturing images of an object, and the optical imaging lens assembly tends to be developed with a compact design and a low cost, while meeting the user requirements for good aberration correction ability, high resolution, and high image quality.

In general, a conventional optical imaging lens assembly of a compact electronic product comes with different designs, including the two-lens, three-lens, four-lens, and five-or-more lens designs. If the image quality of a compact electronic product is taken into consideration, the optical imaging lens assembly with a design of four or five lens elements has advantages on image aberration and modulation transfer function (MTF) performance, wherein the design with five lens elements having a higher resolution, and is suitable for electronic products that require the high quality and high pixel requirements.

In various compact designs of the optical imaging lens assemblies comprised of five lens elements and having a fixed focal length, the prior art adopts different combinations of positive and negative refractive powers, such as a group of stacked lenses as disclosed in U.S. Publications Nos. US2004/0196571 and US2003/0117722 and a combination of a first lens element with negative refractive power and a second lens element with positive refractive power as disclosed in U.S. Pat. No. 7,480,105. Since two glass lenses are cemented, it is difficult to control the cost for mass production, thus incurring a higher manufacturing cost. In addition, the glass lenses generally have a thickness greater than that of plastic lenses, and aspheric lenses cannot be made easily by glass, and the total length of the optical system cannot be shortened easily. If the first lens element with negative refractive power is adopted, the level of difficulty to reduce the total length of the optical system will be much higher.

In products such as compact digital cameras, web cameras, and mobile phone cameras, the optical imaging lens assembly requires a compact design, a short focal length, and a good aberration correction. As disclosed in U.S. Pat. No. 7,502,181, a fourth lens element with negative refractive power and a fifth lens element with negative refractive power are installed, thus it is difficult to reduce the total length of the optical system. In the description above, the conventional designs will increase the rear focal length and the total length of the optical imaging lens assembly. Therefore, the present invention provides a more practical design to shorten the optical imaging lens assembly, while using a combination of refractive powers of five lens elements and a combination of convex and concave optical surfaces, wherein the fourth lens element and the fifth lens element have positive refractive power and negative refractive power respectively, and this complementary combination with a telecentric effect is favorable for reducing the rear focal length and the total length of the optical imaging lens assembly effectively, and further improving the image quality and applying the optical imaging lens assembly to compact electronic products.

SUMMARY OF THE INVENTION

Therefore, it is a primary objective of the present invention to provide an optical imaging lens assembly comprising five non-cemented lenses with refractive power, sequentially arranged from an object side to an image side along an optical axis, comprising: a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element, wherein the first lens element has positive refractive power; the second lens element has negative refractive power; the third lens element with refractive power has a concave object-side surface, and at least one aspheric surface; the fourth lens element with positive refractive power is made of plastic, and has aspheric object-side surface and image-side surface; and the fifth lens element with negative refractive power is made of plastic and has a concave object-side surface and a convex image-side surface, and both object-side surface and image-side surface thereof being aspheric, and the following relations are satisfied:

$$1.55 < f/f_4 < 3.50 \quad (1)$$

$$0.4 < (R_7 + R_8)/(R_7 - R_8) < 2.5 \quad (2)$$

Wherein, f is a focal length of the optical imaging lens assembly, f_4 is a focal length of the fourth lens element, R_7 is a curvature radius of the object-side surface of the fourth lens element, and R_8 is a curvature radius of the image-side surface of the fourth lens element.

On the other hand, the present invention provides an optical imaging lens assembly, as described above, further comprising a stop, wherein the first lens element has a convex object-side surface; the fourth lens element has a convex image-side surface; and the optical imaging lens assembly satisfies one or more of the following relations in addition to the relations (1) and (2):

$$-0.5 < R_9/f < 0 \quad (3)$$

$$0.70 < S_D/T_D < 1.20 \quad (4)$$

$$-3.0 < f/f_5 < -1.7 \quad (5)$$

$$28 < v_1 - v_2 < 45 \quad (6)$$

Wherein, R_9 is a curvature radius of the object-side surface of the fifth lens element, f is the focal length of the optical imaging lens assembly, f_5 is a focal length of the fifth lens element, S_D is an axial distance from the stop to the image-side surface of the fifth lens element, T_D is an axial distance from the object-side surface of the first lens element to the image-side surface of the fifth lens element, v_1 is an Abbe number of the first lens element, and v_2 is an Abbe number of the second lens element.

On the other hand, the present invention provides an optical imaging lens assembly, as described above, further comprising an image sensor at an image plane for imaging an object to be photographed, wherein the first lens element has a convex object-side surface; the fourth lens element has a convex image-side surface; at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point; and the optical imaging lens assembly satisfies one or more of the following relations in addition to the relations (1) and (2):

$$-0.5 < R_9/f < 0 \quad (3)$$

$$0.02 < T_{45}/f < 0.18 \quad (7)$$

$$-10 < v_1 - (v_2 + v_3) < 20 \quad (8)$$

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$$-0.3 < R_9/f < 0 \quad (9)$$

$$-2.0 < (R_9 + R_{10})/(R_9 - R_{10}) < -1.02 \quad (10)$$

$$TTL/ImgH < 1.95 \quad (11)$$

Wherein, R_9 is the curvature radius of the object-side surface of the fifth lens element, R_{10} is a curvature radius of the image-side surface of the fifth lens element, f is the focal length of the optical imaging lens assembly, T_{45} is an axial distance between the fourth lens element and the fifth lens element, v_1 is the Abbe number of the first lens element, v_2 is the Abbe number of the second lens element, v_3 is an Abbe number of the third lens element, TTL is an axial distance from the object-side surface of the first lens element to the image plane, and $ImgH$ is half of the diagonal length of an effective photosensitive area of the image sensor.

On the other hand, the present invention provides an optical imaging lens assembly, as described above, wherein the first lens element has a convex object-side surface; the fourth lens element has a convex image-side surface; and the optical imaging lens assembly satisfies one or more of the following relations in addition to the relations (1) and (2):

$$0.9 < R_8/R_9 < 1.25 \quad (12)$$

$$0.6 < (R_7 + R_8)/(R_7 - R_8) < 1.9 \quad (13)$$

Wherein, R_7 is the curvature radius of the object-side surface of the fourth lens element, R_8 is the curvature radius of the image-side surface of the fourth lens element, and R_9 is the curvature radius of the object-side surface of the fifth lens element.

Another objective of the present invention is to provide an optical imaging lens assembly, comprising five non-cemented lenses with refractive power, sequentially arranged from an object side to an image side along an optical axis, comprising: a first lens element, a second lens element, a third lens element, a fourth lens element, and a fifth lens element, wherein the first lens element has positive refractive power; the second lens element has negative refractive power; the third lens element with negative refractive power has at least one aspheric surface; the fourth lens element with positive refractive power is made of plastic and has both spheric object-side surface and image-side surface; the fifth lens element with negative refractive power is made of plastic and has a concave object-side surface and a convex image-side surface, and both object-side surface and image-side surface thereof being aspheric; and the following relation is satisfied:

$$0.4 < (R_7 + R_8)/(R_7 - R_8) < 2.5 \quad (2)$$

Wherein R_7 is a curvature radius of the object-side surface of the fourth lens element, and R_8 is a curvature radius of the image-side surface of the fourth lens element.

On the other hand, the present invention provides an optical imaging lens assembly, as described above, wherein at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point; and the optical imaging lens assembly satisfies one or more of the following relations in addition to the relation (2):

$$-0.5 < R_9/f < 0 \quad (3)$$

$$1.55 < f/f_4 < 3.50 \quad (4)$$

$$-3.0 < f/f_5 < -1.7 \quad (5)$$

$$-10 < v_1 - (v_2 + v_3) < 20 \quad (8)$$

$$0.9 < R_8/R_9 < 1.25 \quad (12)$$

$$-2.0 < (R_9 + R_{10})/(R_9 - R_{10}) < -1.02 \quad (10)$$

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Wherein, R_8 is the curvature radius of the image-side surface of the fourth lens element, R_9 is a curvature radius of the object-side surface of the fifth lens element, R_{10} is a curvature radius of the image-side surface of the fifth lens element, f is a focal length of the optical imaging lens assembly, f_4 is a focal length of the fourth lens element, f_5 is a focal length of the fifth lens element, v_1 is an Abbe number of the first lens element, v_2 is an Abbe number of the second lens element, and v_3 is an Abbe number of the third lens element.

Another objective of the present invention is to provide an optical imaging lens assembly, comprising five non-cemented lenses with refractive power, sequentially arranged from an object side to an image side along an optical axis, comprising: a first lens element, a second lens element, a third lens element, a fourth lens element, and a fifth lens element; wherein the first lens element has positive refractive power; the second lens element has negative refractive power; the third lens element has refractive power and at least one aspheric surface; the fourth lens element with positive refractive power, is made of plastic and has both aspheric object-side surface and image-side surface; the fifth lens element with negative refractive power is made of plastic and has a concave object-side surface and a convex image-side surface, and both object-side surface and image-side surface thereof being aspheric, and at least one of the object-side surface and image-side surface thereof has at least one inflection point; and the following relation is satisfied:

$$1.55 < f/f_4 < 3.50 \quad (1)$$

$$-0.5 < R_9/f < 0 \quad (3)$$

Wherein, R_9 is a curvature radius of the object-side surface of the fifth lens element, f is a focal length of the optical imaging lens assembly, and f_4 is a focal length of the fourth lens element.

On the other hand, the present invention provides an optical imaging lens assembly, as described above, and satisfies one or more of the following relations in addition to the relations (1) and (3):

$$0.6 < (R_7 + R_8)/(R_7 - R_8) < 1.9 \quad (13)$$

$$0.9 < R_8/R_9 < 1.25 \quad (12)$$

$$1.55 < f/f_4 < 3.50 \quad (1)$$

$$-3.0 < f/f_5 < -1.7 \quad (5)$$

Wherein, R_7 is a curvature radius of the object-side surface of the fourth lens element, R_8 is a curvature radius of the image-side surface of the fourth lens element, R_9 is the curvature radius of the object-side surface of the fifth lens element, f is the focal length of the optical imaging lens assembly, f_4 is the focal length of the fourth lens element, and f_5 is a focal length of the fifth lens element.

With the installation of the aforementioned first lens element, second lens element, third lens element, fourth lens element and fifth lens element with an appropriate interval apart from one another, the present invention can provide a good aberration correction and an advantageous modulation transfer function (MTF) in a greater field of view.

In the optical imaging lens assembly of the present invention, the optical imaging lens assembly is comprised of the first lens element, second lens element, third lens element, fourth lens element and fifth lens element, and the first lens element with positive refractive power provides the refractive power required by the system, and the second lens element with negative refractive power can correct aberrations produced by the lens element with positive refractive power

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effectively and correct the Petzval sum of the system to make the image surface on the edge flatter, and the third lens element can adjust the MTF to correct the aberration of the assembled first lens element and second lens element to enhance the resolution of the optical imaging lens assembly, so that the overall aberration and distortion of the optical imaging lens assembly can meet the high resolution requirement. With the complementary fourth lens element with positive refractive power and fifth lens element with negative refractive power, the telecentric effect can be achieved to facilitate reducing the rear focal length, so as to shorten the total length of the optical imaging lens assembly.

In the optical imaging lens assembly of the present invention, the arrangement of the stop produces a longer distance from the exit pupil of the optical imaging lens assembly to the image plane, so that the light of an image can be projected directly and then received by the image sensor to avoid dark corners or achieve the telecentric effect on the image side. In general, the telecentric effect can improve the brightness of the image plane and enhance the speed of receiving images by the CCD or CMOS image sensor.

In the optical imaging lens assembly of the present invention, the combination of the first lens element with positive refractive power, the second lens element with negative refractive power and the third lens element with positive or negative refractive power, and the mutual compensation of the fourth lens element with positive refractive power and the fifth lens element with negative refractive power can reduce the total length of the optical imaging lens assembly effectively, so that the image sensor can have a larger effective pixel range within the same total length. In other words, a shorter optical imaging lens assembly can be designed with the same effective pixel range of the image sensor.

If the fifth lens element has an inflection point, the inflection point can be used for guiding light of an image with an angle out from the edges of the fifth lens element, such that the light of an image at the off-axis view angle is guided to the image sensor and received by the image sensor. In addition, the fourth lens element and the fifth lens element can be made of plastic to facilitate the manufacture and lower the cost.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1A is a schematic view of an optical lens in accordance with the first preferred embodiment of the present invention;

FIG. 1B is a schematic view of a series of aberration curves of the first preferred embodiment of the present invention;

FIG. 2A is a schematic view of an optical lens in accordance with the second preferred embodiment of the present invention;

FIG. 2B is a schematic view of a series of aberration curves of the second preferred embodiment of the present invention;

FIG. 3A is a schematic view of an optical lens in accordance with the third preferred embodiment of the present invention;

FIG. 3B is a schematic view of a series of aberration curves of the third preferred embodiment of the present invention;

FIG. 4A is a schematic view of an optical lens in accordance with the fourth preferred embodiment of the present invention;

FIG. 4B is a schematic view of a series of aberration curves of the fourth preferred embodiment of the present invention;

FIG. 5A is a schematic view of an optical lens in accordance with the fifth preferred embodiment of the present invention;

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FIG. 5B is a schematic view of a series of aberration curves of the fifth preferred embodiment of the present invention;

FIG. 6A is a schematic view of an optical lens in accordance with the sixth preferred embodiment of the present invention;

FIG. 6B is a schematic view of a series of aberration curves of the sixth preferred embodiment of the present invention;

FIG. 7A is a schematic view of an optical lens in accordance with the seventh preferred embodiment of the present invention;

FIG. 7B is a schematic view of a series of aberration curves of the seventh preferred embodiment of the present invention;

FIG. 8A is a schematic view of an optical lens in accordance with the eighth preferred embodiment of the present invention;

FIG. 8B is a schematic view of a series of aberration curves of the eighth preferred embodiment of the present invention;

FIG. 9A is a schematic view of an optical lens in accordance with the ninth preferred embodiment of the present invention;

FIG. 9B is a schematic view of a series of aberration curves of the ninth preferred embodiment of the present invention.

FIG. 10A is a schematic view of an optical lens in accordance with the tenth preferred embodiment of the present invention; and

FIG. 10B is a schematic view of a series of aberration curves of the tenth preferred embodiment of the present invention.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

With reference to FIG. 1A for an optical imaging lens assembly of the present invention, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, sequentially arranged from an object side to an image side along an optical axis, comprising: a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140 and a fifth lens element 150, wherein the first lens element 110 has positive refractive power; the second lens element 120 has negative refractive power; the third lens element 130 with refractive power has a concave object-side surface 131 and at least one of the object-side surface 131 and an image-side surface 132 thereof being aspheric; the fourth lens element 140 with positive refractive power is made of plastic and has both aspheric object-side surface 141 and image-side surface 142; the fifth lens element 150 with negative refractive power is made of plastic and has a concave object-side surface 151 and a convex image-side surface 152, and both object-side surface 151 and image-side surface 152 thereof being aspheric, and at least one of the object-side surface 151 and image-side surface 152 has at least one inflection point. The optical imaging lens assembly further comprises an aperture stop 100 and an IR-filter 160, wherein the aperture stop 100 is installed between the first lens element 110 and the second lens element 120, and the IR-filter 160 is installed between the fifth lens element 150 and the image plane 170 and generally made of panel glass without affecting the focal length of the optical imaging lens assembly of the present invention. The optical imaging lens assembly further comprises an image sensor 180 installed at an image plane 170 for imaging an object to be photographed. The aspheric surfaces of the first lens element 110, second lens element 120, third lens element 130, fourth lens element 140 and fifth lens element 150 comply with the aspherical surface formula as given in Equation (14).

$$X(Y) = \frac{(Y^2/R)}{1 + \sqrt{1 - (1 + K)(Y/R)^2}} + \sum_i (A_i) \cdot (Y^i) \quad (14)$$

Wherein,

X is the relative height from a point on the spherical surface with a distance Y from the optical axis to a tangent plane at the tip of the optical axis of the spherical surface;

Y is the distance between a point on the curve of the spherical surface and the optical axis;

R is the curvature radius near the optical axis;

K is the conic coefficient; and

A_i is the i^{th} level aspherical surface coefficient.

In the optical imaging lens assembly of the present invention, the first lens element **110**, second lens element **120**, third lens element **130**, fourth lens element **140** and fifth lens element **150** can be spherical or aspherical surfaces. and if aspheric optical surfaces are adopted, then the curvature radius of the optical surface can be used for changing the refractive power to reduce or eliminate aberrations, so as to decrease the number of lens elements used in the optical imaging lens assembly and shorten the total length of the optical imaging lens assembly effectively. With the installation of the first lens element **110**, second lens element **120**, third lens element **130**, fourth lens element **140** and fifth lens element **150**, the optical imaging lens assembly of the present invention satisfies the relations (1) and (2).

In the optical imaging lens assembly of the present invention, the main positive refractive power is provided by the first lens element **110** and the fourth lens element **140**. If the relation (1) is satisfied, the ratio of the focal length f_4 of the fourth lens element **140** to the focal length f of the optical imaging lens assembly is limited to allocate the refractive power required by the fourth lens element **140** of the optical imaging lens assembly effectively, so as to reduce the system sensitivity and provide appropriate positive refractive power required by the system. In addition, the fourth lens element **140** has a convex image-side surface **142**, such that when the ratio of curvature radii of the object-side surface **141** and the image-side surface **142** as given in the relation (2) is limited, the larger the curvature radius of the image-side surface **142**, the smaller is the refractive power of the fourth lens element **140**. Therefore, the fourth lens element **140** can be adjusted to have appropriate refractive power to reduce the system sensitivity and facilitate the manufacture, so as to improve the yield rate and the production cost.

If the relations (3) and (9) are satisfied, the negative refractive power will be relatively weaker when the ratio in these relations is too large since the fifth lens element **150** has a concave object-side surface **151**, so that the aberration correction ability is weakened. If the ratio in these relations is too small, the negative refractive power will be relatively stronger, and the total length cannot be shortened effectively. Therefore, these relations can be limited within an appropriate range to achieve both effects of correcting the aberration and reducing the total length effectively. Similarly, if the relation (12) is satisfied, the refractive powers of the fourth lens element **140** and the fifth lens element **150** can be allocated or adjusted appropriately to further correct the aberration of the optical imaging lens assembly and adjust the rear focal length and the total length. If the curvature radius R_9 of the object-side surface **151** of the fifth lens element **150** and the curvature radius R_{10} of the image-side surface **152** of the fifth lens element **150** are limited (as shown in the relation (10)), the change of the shape of the fifth lens element **150** can be limited, not only facilitating the correction of the system aberration, but also facilitating the allocation of the refractive

power of the fifth lens element **150** and complement the positive refractive power of the fourth lens element **140**.

If the relation (11) is satisfied, the total length of the optical imaging lens assembly can be reduced effectively, so that the image sensor can have a larger effective pixel range within the same total length. Similarly, if the relation (4) is satisfied, the position of the stop and the distance from the first lens element **110** to the fifth lens element **150** can be adjusted to shorten the length of the optical imaging lens assembly. If the relation (7) is satisfied, the focal length f per unit length of the optical imaging lens assembly can adjust the distance from the image-side surface **142** of the fourth lens element **140** to the object-side surface **151** of the fifth lens element **150** to facilitate reducing the total length of the optical imaging lens assembly.

If the relation (6) is satisfied, the difference between the Abbe number v_1 of the first lens element **110** and the Abbe number v_2 of the second lens element **120** falls within an appropriate range to correct the chromatic aberration produced by the first lens element **110** and the second lens element **120** and improve the chromatic aberration correction ability of the second lens element **120**. Similarly, if the relation (8) is satisfied, the chromatic aberration correction ability of the second lens element **120** and the third lens element **130** can be enhanced.

The optical imaging lens assembly of the present invention is described by means of preferred embodiments with relevant drawings as follows.

<First Preferred Embodiment>

With reference to FIGS. 1A and 1B for a schematic view and a series of aberration curves of an optical image lens assembly in accordance with the first preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, an aperture stop **100** and an IR-filter **160**. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: a plastic first lens element **110** with positive refractive power, having a convex object-side surface **111** and a convex image-side surface **112**, and both object-side surface **111** and image-side surface **112** thereof being aspheric; an aperture stop **100**; a plastic second lens element **120** with negative refractive power, having a convex object-side surface **121** and a concave image-side surface **122**, and both object-side surface **121** and image-side surface **122** thereof being aspheric; a plastic third lens element **130** with negative refractive power, having a concave object-side surface **131** and a convex image-side surface **132**, and both object-side surface **131** and image-side surface **132** being aspheric; a plastic fourth lens element **140** with positive refractive power, having a convex object-side surface **141** and a convex image-side surface **142**, and both object-side surface **141** and image-side surface **142** being aspheric; a plastic fifth lens element **150** with negative refractive power, having a concave object-side surface **151** and a convex image-side surface **152**, and both object-side surface **151** and image-side surface **152** thereof being aspheric, and at least one of the object-side surface **151** and image-side surface **152** thereof having at least one inflection point; a glass IR-filter **160**, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor **180** installed at an image plane **170**. With the combination of the five lens elements, the aperture stop **100** and the IR-filter **160**, an image of the object to be photographed can be formed at the image sensor **180**.

TABLE 1

Optical data of this preferred embodiment f = 4.14 mm, Fno = 2.40, HFOV = 34.5 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.736030 (ASP)	0.631	Plastic	1.544	55.9	2.95
2		-18.509600 (ASP)	0.038				
3	Ape. Stop	Plano	0.008				
4	Lens 2	4.076600 (ASP)	0.264	Plastic	1.634	23.8	-5.62
5		1.852670 (ASP)	0.602				
6	Lens 3	-3.812000 (ASP)	0.287	Plastic	1.634	23.8	-6.56
7		-47.393400 (ASP)	0.252				
8	Lens 4	12.987400 (ASP)	1.065	Plastic	1.544	55.9	1.86
9		-1.065340 (ASP)	0.510				
10	Lens 5	-1.063820 (ASP)	0.345	Plastic	1.544	55.9	-2.09
11		-18.450200 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	—
13		Plano	0.335				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 1, wherein the object-side surface and the image-side surface of the first lens element **110** to the fifth lens element **150** comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 2 as follows:

TABLE 2

Aspheric coefficients of this preferred embodiment					
	Surface #				
	1	2	4	5	6
k =	-7.26016E-01	-1.00000E+00	-7.50656E+00	-1.00000E+00	3.00000E+00
A4 =	1.73325E-02	8.42251E-03	-7.85854E-02	-1.02422E-01	-2.69651E-01
A6 =	2.43582E-02	5.18034E-02	1.85765E-01	1.41611E-01	8.93336E-02
A8 =	-4.91007E-02	-5.35781E-02	-1.89019E-01	-8.06089E-02	-1.75391E-01
A10 =	5.50506E-02	-6.57263E-02	1.28560E-01	1.82684E-02	1.87723E-01
A12 =	-3.34017E-02	1.19766E-01	-1.05664E-01	3.10189E-02	-2.52004E-02
A14 =	5.21552E-03	-4.98156E-02	8.44373E-02	-5.02247E-04	2.49936E-04
	Surface #				
	7	8	9	10	11
k =	-1.00000E+00	-1.00000E+00	-7.47279E-01	-2.13568E+00	-1.00000E+00
A4 =	-2.73081E-01	-1.39149E-01	9.05801E-02	1.15389E-01	3.03435E-02
A6 =	1.70993E-01	7.38942E-02	-3.22101E-02	-4.26279E-02	-1.60039E-02
A8 =	-1.28326E-01	9.50584E-03	3.62534E-02	7.52812E-03	1.96761E-03
A10 =	8.02572E-02	-1.93390E-02	-1.08500E-02	-6.83537E-04	1.70253E-05
A12 =	-3.12027E-03	6.44817E-03	1.20830E-03	5.17745E-05	-2.00616E-05
A14 =	-5.82394E-03	-7.68452E-04	-3.43090E-05	-3.81357E-06	-1.72308E-06
A16 =				6.92412E-08	4.07827E-07

With reference to Table 1 and FIG. 1B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.14$ (mm), an overall aperture value (f-number) $Fno=2.40$, and a half of maximum view angle $HFOV=34.5^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 3 below, and the related symbols have been described above and thus will not be described again.

TABLE 3

Data of related relations of this preferred embodiment			
Relation	Data	Relation	Data
R_8/R_9	1.00	T_{45}/f	0.12
$v_1 - v_2$	32.1	f/f_5	-1.98
$v_1 - (v_2 + v_3)$	8.3	f/f_4	2.23
R_9/f	-0.26	S_D/T_D	0.83
$(R_7 + R_8)/(R_7 - R_8)$	0.85	$TTL/ImgH$	1.80
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.12		

According to the optical data as shown in Table 1 and the aberration curves as shown in FIG. 1B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

<Second Preferred Embodiment>

With reference to FIGS. 2A and 2B for a schematic view and a series of aberration curves of an optical image lens

positive refractive power, having a convex object-side surface **211** and a convex image-side surface **212**, and both object-side surface **211** and image-side surface **212** thereof being aspheric; an aperture stop **200**; a plastic second lens element **220** with negative refractive power, having a concave object-side surface **221** and a concave image-side surface **222**, and both object-side surface **221** and image-side surface **222** thereof being aspheric; a plastic third lens element **230** with negative refractive power, having a concave object-side surface **231** and a convex image-side surface **232**, and both object-side surface **231** and image-side surface **232** being aspheric; a plastic fourth lens element **240** with positive refractive power, having a convex object-side surface **241** and a convex image-side surface **242**, and both object-side surface **241** and image-side surface **242** being aspheric; a plastic fifth lens element **250** with negative refractive power, having a concave object-side surface **251** and a convex image-side surface **252**, and both object-side surface **251** and image-side surface **252** thereof being aspheric, and at least one of the object-side surface **251** and image-side surface **252** thereof having at least one inflection point; a glass IR-filter **260**, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor **280** installed at an image plane **270**. With the combination of the five lens elements, the aperture stop **200** and the IR-filter **260**, an image of the object to be photographed can be formed at the image sensor **280**.

TABLE 4

Optical data of this preferred embodiment $f = 4.16$ mm, $Fno = 2.42$, $HFOV = 34.6$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.743580 (ASP)	0.651	Plastic	1.544	55.9	2.92
2		-15.503700 (ASP)	0.024				
3	Ape. Stop	Plano	0.070				
4	Lens 2	-19.011400 (ASP)	0.264	Plastic	1.634	23.8	-7.13
5		5.958400 (ASP)	0.528				
6	Lens 3	-2.616620 (ASP)	0.347	Plastic	1.634	23.8	-5.08
7		-14.694800 (ASP)	0.292				
8	Lens 4	7.241000 (ASP)	1.062	Plastic	1.544	55.9	1.81
9		-1.078390 (ASP)	0.497				
10	Lens 5	-0.988750 (ASP)	0.345	Plastic	1.544	55.9	-1.93
11		-18.450200 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	—
13		Plano	0.334				
14	Image	Plano	—				

Note:
Reference wavelength is 587.6 nm.
ASP stands for aspherical surfaces.

assembly in accordance with the second preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, an aperture stop **200** and an IR-filter **260**. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: a plastic first lens element **210** with

The optical data of this preferred embodiment are listed in Table 4, wherein the object-side surface and the image-side surface of the first lens element **210** to the fifth lens element **250** comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 5 as follows:

TABLE 5

Aspheric coefficients of this preferred embodiment					
Surface #					
	1	2	4	5	6
k =	-8.04791E-01	-1.00000E+00	-1.00000E+01	-1.00000E+00	2.99994E+00
A4 =	1.50510E-02	-6.12298E-03	-5.50116E-02	-8.91182E-02	-3.09675E-01
A6 =	3.06131E-02	4.82198E-02	1.77356E-01	8.72488E-02	6.47383E-02
A8 =	-5.40847E-02	-5.15160E-02	-2.83564E-01	-1.41840E-01	-1.72737E-01
A10 =	5.53797E-02	-5.85889E-02	2.15582E-01	3.06101E-02	2.20109E-01
A12 =	-3.36573E-02	1.18791E-01	-1.07113E-01	3.27214E-02	-2.50722E-02
A14 =	5.15173E-03	-5.35846E-02	8.38942E-02	4.06113E-03	5.01595E-04
Surface #					
	7	8	9	10	11
k =	-1.00000E+00	-1.00000E+00	-7.57505E-01	-1.96020E+00	-1.00000E+00
A4 =	-2.90351E-01	-1.42970E-01	9.22654E-02	1.16560E-01	3.15738E-02
A6 =	1.86760E-01	7.29719E-02	-3.20087E-02	-4.24710E-02	-1.60778E-02
A8 =	-1.16970E-01	9.44155E-03	3.61147E-02	7.57437E-03	2.03632E-03
A10 =	8.04911E-02	-1.92238E-02	-1.09229E-02	-6.75179E-04	1.62791E-05
A12 =	-4.26514E-03	6.45602E-03	1.20034E-03	5.24733E-05	-2.09779E-05
A14 =	-5.87485E-03	-7.78351E-04	-3.41938E-05	-4.13325E-06	-1.81226E-06
A16 =				9.10163E-09	4.02385E-07

With reference to Table 4 and FIG. 2B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.16$ (mm), an overall aperture value (f-number) $Fno=2.42$, and a half of maximum view angle $HFOV=34.6^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 6 below, and the related symbols have been described above and thus will not be described again.

TABLE 6

Data of related relations of this preferred embodiment			
Relation	Data	Relation	Data
R_8/R_9	1.09	T_{45}/f	0.12
$v_1 - v_2$	32.1	f/f_3	-2.16
$v_1 - (v_2 + v_3)$	8.3	f/f_4	2.30
R_9/f	-0.24	S_D/T_D	0.83
$(R_7 + R_8)/(R_7 - R_8)$	0.74	$TTL/ImgH$	1.83
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.11		

According to the optical data as shown in Table 4 and the aberration curves as shown in FIG. 2B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

<Third Preferred Embodiment>

With reference to FIGS. 3A and 3B for a schematic view and a series of aberration curves of an optical image lens assembly in accordance with the third preferred embodiment of the present invention respectively, the optical imaging lens

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assembly comprises five non-cemented lenses with refractive power, an aperture stop **300** and an IR-filter **360**. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: a plastic first lens element **310** with positive refractive power, having a convex object-side surface **311** and a convex image-side surface **312**, and both object-side surface **311** and image-side surface **312** thereof being aspheric; an aperture stop **300**; a plastic second lens element **320** with negative refractive power, having a concave object-side surface **321** and a convex image-side surface **322**, and both object-side surface **321** and image-side surface **322** thereof being aspheric; a plastic third lens element **330** with negative refractive power, having a concave object-side surface **331** and a convex image-side surface **332**, and both object-side surface **331** and image-side surface **332** being aspheric; a plastic fourth lens element **340** with positive refractive power, having a convex object-side surface **341** and a convex image-side surface **342**, and both object-side surface **341** and image-side surface **342** being aspheric; a plastic fifth lens element **350** with negative refractive power, having a concave object-side surface **351** and a convex image-side surface **352**, and both object-side surface **351** and image-side surface **352** thereof being aspheric, and at least one of the object-side surface **351** and image-side surface **352** thereof having at least one inflection point; a glass IR-filter **360**, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor **380** installed at an image plane **370**. With the combination of the five lens elements, the aperture stop **300** and the IR-filter **360**, an image of the object to be photographed can be formed at the image sensor **380**.

TABLE 7

Optical data of this preferred embodiment $f = 4.17$ mm, $Fno = 2.44$, $HFOV = 34.6$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.749820 (ASP)	0.624	Plastic	1.544	55.9	2.91
2		-14.779100 (ASP)	0.009				
3	Ape. Stop	Plano	0.115				

TABLE 7-continued

Optical data of this preferred embodiment $f = 4.17$ mm, $Fno = 2.44$, $HFOV = 34.6$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
4	Lens 2	-4.390800 (ASP)	0.264	Plastic	1.634	23.8	-9.11
5		-18.761700 (ASP)	0.498				
6	Lens 3	-2.317490 (ASP)	0.377	Plastic	1.634	23.8	-4.35
7		-15.393300 (ASP)	0.306				
8	Lens 4	6.249000 (ASP)	1.044	Plastic	1.544	55.9	1.83
9		-1.114690 (ASP)	0.520				
10	Lens 5	-0.993840 (ASP)	0.345	Plastic	1.544	55.9	-1.94
11		-18.450200 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.333				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

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The optical data of this preferred embodiment are listed in Table 7, wherein the object-side surface and the image-side surface of the first lens element **310** to the fifth lens element **350** comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 8 as follows:

TABLE 8

Aspheric coefficients of this preferred embodiment					
	Surface #				
	1	2	4	5	6
k =	-9.13257E-01	-1.00000E+00	-1.00000E+01	-1.00000E+00	3.00000E+00
A4 =	1.27045E-02	-2.06057E-02	-2.61372E-02	-5.10255E-02	-3.17271E-01
A6 =	2.74654E-02	4.93646E-02	1.61260E-01	6.96161E-02	7.92072E-02
A8 =	-5.04598E-02	-7.10990E-02	-3.05703E-01	-1.51333E-01	-1.06057E-01
A10 =	4.25743E-02	-4.72814E-02	2.36214E-01	3.38989E-02	1.63110E-01
A12 =	-3.36572E-02	1.18791E-01	-1.07113E-01	3.27215E-02	-2.50721E-02
A14 =	5.15176E-03	-5.35846E-02	8.38942E-02	4.06116E-03	5.01624E-04
	Surface #				
	7	8	9	10	11
k =	-1.00000E+00	-1.00000E+00	-7.64055E-01	-2.00233E+00	-1.00000E+00
A4 =	-3.00743E-01	-1.40069E-01	9.05594E-02	1.13331E-01	3.46905E-02
A6 =	2.10175E-01	7.09015E-02	-3.18076E-02	-4.23894E-02	-1.64984E-02
A8 =	-1.19082E-01	9.38216E-03	3.56283E-02	7.60914E-03	2.07772E-03
A10 =	7.13337E-02	-1.91191E-02	-1.10176E-02	-6.72670E-04	1.74151E-05
A12 =	-6.43791E-03	6.46984E-03	1.19316E-03	5.38167E-05	-2.14689E-05
A14 =	-3.75455E-03	-7.85642E-04	-2.86394E-05	-4.10511E-06	-1.76468E-06
A16 =				-3.29942E-08	4.02907E-07

With reference to Table 7 and FIG. 3B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.17$ (mm), an overall aperture value (f-number) $Fno=2.44$, and a half of maximum view angle $HFOV=34.6^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 9 below, and the related symbols have been described above and thus will not be described again.

TABLE 9

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	1.12
$v_1 - v_2$	32.1

TABLE 9-continued

Data of related relations of this preferred embodiment	
Relation	Data
$v_1 - (v_2 + v_3)$	8.3
R_9/f	-0.24
$(R_7 + R_8)/(R_7 - R_8)$	0.70
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.11
T_{45}/f	0.12
f/f_5	-2.15
f/f_4	2.28
S_D/T_D	0.85
$TTL/ImgH$	1.83

According to the optical data as shown in Table 7 and the aberration curves as shown in FIG. 3B, the optical imaging

lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

<Fourth Preferred Embodiment>

With reference to FIGS. 4A and 4B for a schematic view and a series of aberration curves of an optical image lens assembly in accordance with the fourth preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, an aperture stop 400 and an IR-filter 460. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: a plastic first lens element 410 with positive refractive power, having a convex object-side surface 411 and a convex image-side surface 412, and both object-side surface 411 and image-side surface 412 thereof being aspheric; an aperture stop 400; a plastic second lens element 420 with negative refractive power, having a concave object-side surface 421 and a concave image-side surface 422, and

both object-side surface 421 and image-side surface 422 thereof being aspheric; a plastic third lens element 430 with negative refractive power, having a concave object-side surface 431 and a convex image-side surface 432, and both object-side surface 431 and image-side surface 432 being aspheric; a plastic fourth lens element 440 with positive refractive power, having a concave object-side surface 441 and a convex image-side surface 442, and both object-side surface 441 and image-side surface 442 being aspheric; a plastic fifth lens element 450 with negative refractive power, having a concave object-side surface 451 and a convex image-side surface 452, and both object-side surface 451 and image-side surface 452 thereof being aspheric, and at least one of the object-side surface 451 and image-side surface 452 thereof having at least one inflection point; a glass IR-filter 460, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor 480 installed at an image plane 470. With the combination of the five lens elements, the aperture stop 400 and the IR-filter 460, an image of the object to be photographed can be formed at the image sensor 480.

TABLE 10

Optical data of this preferred embodiment f = 4.57 mm, Fno = 2.45, HFOV = 32.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.558140 (ASP)	0.757	Plastic	1.544	55.9	2.73
2		-25.730400 (ASP)	0.046				
3	Ape. Stop	Plano	0.040				
4	Lens 2	-22.159700 (ASP)	0.260	Plastic	1.634	23.8	-4.51
5		3.291400 (ASP)	0.458				
6	Lens 3	-5.429400 (ASP)	0.315	Plastic	1.634	23.8	-22.89
7		-8.879700 (ASP)	0.444				
8	Lens 4	-6.827600 (ASP)	0.824	Plastic	1.544	55.9	2.42
9		-1.150360 (ASP)	0.506				
10	Lens 5	-1.133390 (ASP)	0.350	Plastic	1.535	56.3	-2.27
11		-18.587400 (ASP)	0.700				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.278				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 10, wherein the object-side surface and the image-side surface of the first lens element 410 to the fifth lens element 450 comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 11 as follows:

TABLE 11

Aspheric coefficients of this preferred embodiment					
Surface #	1	2	4	5	6
k =	-5.82989E-01	-1.00000E+00	3.73983E+02	-1.00000E+00	-6.01011E+01
A4 =	2.19235E-02	4.59412E-02	4.77999E-02	1.25562E-02	-2.78959E-01
A6 =	2.66709E-02	-3.39737E-02	-2.09552E-02	-3.54047E-03	5.08314E-02
A8 =	-3.95824E-02	4.62463E-02	-3.56539E-04	3.68894E-02	-1.68151E-01
A10 =	5.72844E-02	-4.28087E-02	8.18099E-02	-6.79630E-02	2.03302E-02
A12 =	-3.52738E-02	2.03949E-03	-1.40835E-01	5.35561E-02	8.46118E-02
A14 =	8.60352E-03	1.36219E-02	8.41915E-02	7.93322E-09	7.90905E-09
Surface #	7	8	9	10	11
k =	-1.00000E+00	-2.08437E+00	-8.17755E-01	-1.86980E+00	-1.00000E+00
A4 =	-1.91839E-01	-1.01153E-01	5.76715E-02	1.13165E-01	1.98571E-02
A6 =	6.55814E-02	4.43497E-02	-2.03799E-02	-4.25816E-02	-1.43392E-02

TABLE 11-continued

Aspheric coefficients of this preferred embodiment					
A8 =	-9.40251E-02	9.23350E-03	3.56954E-02	7.50418E-03	2.03738E-03
A10 =	6.95807E-02	-1.39739E-02	-1.23582E-02	-6.76709E-04	5.61727E-06
A12 =	-1.09319E-02	7.14900E-03	8.56881E-04	5.55063E-05	-2.96278E-05
A14 =	1.17069E-02	-2.14174E-03	2.75189E-05	-3.13819E-06	-2.40407E-06
A16 =				5.63188E-08	8.24025E-07

With reference to Table 10 and FIG. 4B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.57$ (mm), an overall aperture value (f-number) $Fno=2.45$, and a half of maximum view angle $HFOV=32.1^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 12 below, and the related symbols have been described above and thus will not be described again.

TABLE 12

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	1.01
$v_1 - v_2$	32.5
$v_1 - (v_2 + v_3)$	9.1
R_9/f	-0.25
$(R_7 + R_8)/(R_7 - R_8)$	1.41
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.13
T_{45}/f	0.11
f/f_3	-2.01
f/f_4	1.89
S_D/T_D	0.80
$TTL/lmgH$	1.81

According to the optical data as shown in Table 10 and the aberration curves as shown in FIG. 4B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

<Fifth Preferred Embodiment>

With reference to FIGS. 5A and 5B for a schematic view and a series of aberration curves of an optical image lens

assembly in accordance with the fifth preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, an aperture stop 500 and an IR-filter 560. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: a plastic first lens element 510 with positive refractive power, having a convex object-side surface 511 and a convex image-side surface 512, and both object-side surface 511 and image-side surface 512 thereof being aspheric; an aperture stop 500; a plastic second lens element 520 with negative refractive power, having a concave object-side surface 521 and a concave image-side surface 522, and both object-side surface 521 and image-side surface 522 thereof being aspheric; a plastic third lens element 530 with positive refractive power, having a concave object-side surface 531 and a convex image-side surface 532, and both object-side surface 531 and image-side surface 532 being aspheric; a plastic fourth lens element 540 with positive refractive power, having a concave object-side surface 541 and a convex image-side surface 542, and both object-side surface 541 and image-side surface 542 being aspheric; a plastic fifth lens element 550 with negative refractive power, having a concave object-side surface 551 and a convex image-side surface 552, and both object-side surface 551 and image-side surface 552 thereof being aspheric, and at least one of the object-side surface 551 and image-side surface 552 thereof having at least one inflection point; a glass IR-filter 560, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor 580 installed at an image plane 570. With the combination of the five lens elements, the aperture stop 500 and the IR-filter 560, an image of the object to be photographed can be formed at the image sensor 580.

TABLE 13

Optical data of this preferred embodiment $f = 4.50$ mm, $Fno = 2.45$, $HFOV = 32.6$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.596690 (ASP)	0.748	Plastic	1.544	55.9	2.62
2		-11.304400 (ASP)	0.040				
3	Ape. Stop	Plano	0.067				
4	Lens 2	-15.750900 (ASP)	0.260	Plastic	1.634	23.8	-3.62
5		2.706830 (ASP)	0.398				
6	Lens 3	-19.531200 (ASP)	0.303	Plastic	1.634	23.8	33.91
7		-10.295300 (ASP)	0.565				
8	Lens 4	-4.707900 (ASP)	0.765	Plastic	1.544	55.9	2.89
9		-1.247370 (ASP)	0.610				
10	Lens 5	-1.166940 (ASP)	0.320	Plastic	1.544	55.9	-2.37
11		-13.605400 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.182				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

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The optical data of this preferred embodiment are listed in Table 13, wherein the object-side surface and the image-side surface of the first lens element **510** to the fifth lens element **550** comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 14 as follows:

TABLE 14

Aspheric coefficients of this preferred embodiment					
Surface #	1	2	4	5	6
k =	-6.38905E-01	-1.00000E+00	3.00000E+00	-1.00000E+00	-4.00000E+01
A4 =	1.98831E-02	5.58766E-02	4.97593E-02	1.37048E-03	-1.96079E-01
A6 =	2.26866E-02	-3.30644E-02	-4.89253E-03	3.50307E-02	2.20981E-02
A8 =	-3.85164E-02	2.85439E-02	-2.45694E-02	7.96323E-03	-1.39082E-01
A10 =	5.41141E-02	-4.06475E-02	9.64193E-02	-4.57772E-02	5.22288E-02
A12 =	-3.53269E-02	3.86308E-02	-1.39896E-01	5.34506E-02	2.56772E-02
A14 =	8.81495E-03	-1.39844E-02	8.44376E-02	-5.02024E-04	2.50159E-04
Surface #	7	8	9	10	11
k =	-1.00000E+00	-8.21067E+00	-7.41870E-01	-1.64207E+00	-1.00000E+00
A4 =	-1.49086E-01	-9.43368E-02	3.53659E-02	1.14770E-01	2.54914E-02
A6 =	4.29332E-02	4.63557E-02	-1.01167E-02	-4.22375E-02	-1.51341E-02
A8 =	-8.21071E-02	1.25029E-02	3.31653E-02	7.53931E-03	1.99340E-03
A10 =	6.42961E-02	-1.70870E-02	-1.21002E-02	-6.77923E-04	1.58247E-05
A12 =	-2.17038E-02	6.63112E-03	1.04455E-03	5.42692E-05	-2.62516E-05
A14 =	1.56627E-02	-1.29311E-03	2.28381E-05	-3.44482E-06	-2.31161E-06
A16 =				7.42044E-09	7.01646E-07

With reference to Table 13 and FIG. 5B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.50$ (mm), an overall aperture value (f-number) $Fno=2.45$, and a half of maximum view angle $HFOV=32.6^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 15 below, and the related symbols have been described above and thus will not be described again.

TABLE 15

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	1.07
$v_1 - v_2$	32.1
$v_1 - (v_2 + v_3)$	8.3
R_9/f	-0.26
$(R_7 + R_8)/(R_7 - R_8)$	1.72
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.19
T_{45}/f	0.14
f/I_3	-1.90
f/I_4	1.56
S_D/T_D	0.81
$TTL/ImgH$	1.77

According to the optical data as shown in Table 13 and the aberration curves as shown in FIG. 5B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

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<Sixth Preferred Embodiment>

With reference to FIGS. 6A and 6B for a schematic view and a series of aberration curves of an optical image lens assembly in accordance with the sixth preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive

power, an aperture stop **600** and an IR-filter **660**. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: a plastic first lens element **610** with positive refractive power, having a convex object-side surface **611** and a concave image-side surface **612**, and both object-side surface **611** and image-side surface **612** thereof being aspheric; an aperture stop **600**; a plastic second lens element **620** with negative refractive power, having a convex object-side surface **621** and a concave image-side surface **622**, and both object-side surface **621** and image-side surface **622** thereof being aspheric; a plastic third lens element **630** with negative refractive power, having a concave object-side surface **631** and a convex image-side surface **632**, and both object-side surface **631** and image-side surface **632** being aspheric; a plastic fourth lens element **640** with positive refractive power, having a concave object-side surface **641** and a convex image-side surface **642**, and both object-side surface **641** and image-side surface **642** being aspheric; a plastic fifth lens element **650** with negative refractive power, having a concave object-side surface **651** and a convex image-side surface **652**, and both object-side surface **651** and image-side surface **652** thereof being aspheric, and at least one of the object-side surface **651** and image-side surface **652** thereof having at least one inflection point; a glass IR-filter **660**, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor **680** installed at an image plane **670**. With the combination of the five lens elements, the aperture stop **600** and the IR-filter **660**, an image of the object to be photographed can be formed at the image sensor **680**.

TABLE 16

Optical data of this preferred embodiment f = 4.52 mm, Fno = 2.45, HFOV = 32.6 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.622300 (ASP)	0.761	Plastic	1.544	55.9	3.21
2		19.193900 (ASP)	0.074				
3	Ape. Stop	Plano	0.022				
4	Lens 2	7.294400 (ASP)	0.272	Plastic	1.634	23.8	-5.74
5		2.393730 (ASP)	0.541				
6	Lens 3	-7.962000 (ASP)	0.315	Plastic	1.634	23.8	-14.01
7		-78.153000 (ASP)	0.265				
8	Lens 4	-22.202900 (ASP)	0.979	Plastic	1.544	55.9	2.32
9		-1.212880 (ASP)	0.588				
10	Lens 5	-1.150240 (ASP)	0.335	Plastic	1.544	55.9	-2.36
11		-11.947400 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.332				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 16, wherein the object-side surface and the image-side surface of the first lens element **610** to the fifth lens element

650 comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 17 as follows:

TABLE 17

Aspheric coefficients of this preferred embodiment					
Surface #	1	2	4	5	6
k =	-6.16641E-01	-1.00000E+00	-4.00000E+01	-1.00000E+00	-3.08479E+01
A4 =	2.09194E-02	-2.82463E-02	-8.44754E-02	-5.66286E-02	-2.19555E-01
A6 =	2.57408E-02	5.82004E-02	1.35747E-01	9.98658E-02	3.27024E-02
A8 =	-3.76497E-02	-5.55134E-03	-8.27094E-02	-9.26200E-03	-9.91244E-02
A10 =	5.21943E-02	-7.12587E-02	3.91735E-02	-7.27534E-02	4.32878E-02
A12 =	-3.18382E-02	8.13985E-02	-9.45931E-02	6.97965E-02	1.48318E-02
A14 =	8.79930E-03	-2.98052E-02	8.44375E-02	-5.02081E-04	2.50102E-04
Surface #	7	8	9	10	11
k =	-1.00000E+00	9.64855E+01	-7.35385E-01	-1.60695E+00	-1.00000E+00
A4 =	-2.08737E-01	-1.03421E-01	5.23187E-02	1.16650E-01	2.95461E-02
A6 =	1.09074E-01	6.41224E-02	-1.84533E-02	-4.28065E-02	-1.49827E-02
A8 =	-1.06581E-01	1.06051E-02	3.26842E-02	7.43240E-03	1.90029E-03
A10 =	6.97418E-02	-1.93391E-02	-1.12445E-02	-6.83192E-04	1.11980E-05
A12 =	-1.68393E-02	6.34895E-03	1.22385E-03	5.58375E-05	-2.10342E-05
A14 =	2.57519E-03	-7.79704E-04	-2.45747E-05	-2.93506E-06	-1.78156E-06
A16 =				1.23844E-08	4.82490E-07

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With reference to Table 16 and FIG. 6B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.52$ (mm), an overall aperture value (f-number) $Fno=2.45$, and a half of maximum view angle $HFOV=32.6^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 18 below, and the related symbols have been described above and thus will not be described again.

TABLE 18

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	1.05
$v_1 - v_2$	32.1
$v_1 - (v_2 + v_3)$	8.3
R_9/f	-0.26
$(R_7 + R_8)/(R_7 - R_8)$	1.12
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.21
T_{45}/f	0.13
f/I_5	-1.92
f/I_4	1.95
S_D/T_D	0.80
$TTL/ImgH$	1.85

According to the optical data as shown in Table 16 and the aberration curves as shown in FIG. 6B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

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refractive power, an aperture stop **700** and an IR-filter **760**. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: an aperture stop **700**; a plastic first lens element **710** with positive refractive power, having a convex object-side surface **711** and a concave image-side surface **712**, and both object-side surface **711** and image-side surface **712** thereof being aspheric; a plastic second lens element **720** with negative refractive power, having a concave object-side surface **721** and a concave image-side surface **722**, and both object-side surface **721** and image-side surface **722** thereof being aspheric; a plastic third lens element **730** with negative refractive power, having a concave object-side surface **731** and a convex image-side surface **732**, and both object-side surface **731** and image-side surface **732** being aspheric; a plastic fourth lens element **740** with positive refractive power, having a concave object-side surface **741** and a convex image-side surface **742**, and both object-side surface **741** and image-side surface **742** being aspheric; a plastic fifth lens element **750** with negative refractive power, having a concave object-side surface **751** and a convex image-side surface **752**, and both object-side surface **751** and image-side surface **752** thereof being aspheric, and at least one of the object-side surface **751** and image-side surface **752** thereof having at least one inflection point; a glass IR-filter **760**, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor **780** installed at an image plane **770**. With the combination of the five lens elements, the aperture stop **700** and the IR-filter **760**, an image of the object to be photographed can be formed at the image sensor **780**.

TABLE 19

Optical data of this preferred embodiment $f = 4.20$ mm, $Fno = 2.40$, $HFOV = 34.1$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.280				
2	Lens 1	1.505740 (ASP)	0.582	Plastic	1.544	55.9	2.93
3		24.096400 (ASP)	0.109				
4	Lens 2	-18.532300 (ASP)	0.258	Plastic	1.634	23.8	-6.19
5		5.002600 (ASP)	0.527				
6	Lens 3	-5.066300 (ASP)	0.296	Plastic	1.634	23.8	-9.07
7		-43.478300 (ASP)	0.294				
8	Lens 4	-168.891600 (ASP)	0.878	Plastic	1.544	55.9	2.20
9		-1.189250 (ASP)	0.600				
10	Lens 5	-1.058970 (ASP)	0.352	Plastic	1.544	55.9	-2.17
11		-11.454800 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.234				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

<Seventh Preferred Embodiment>

With reference to FIGS. 7A and 7B for a schematic view and a series of aberration curves of an optical image lens assembly in accordance with the seventh preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with

The optical data of this preferred embodiment are listed in Table 19, wherein the object-side surface and the image-side surface of the first lens element **710** to the fifth lens element **750** comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 20 as follows:

TABLE 20

Aspheric coefficients of this preferred embodiment					
Surface #	2	3	4	5	6
k =	-4.69360E-01	-1.00000E+00	3.00000E+00	-1.00000E+00	-1.04289E+01
A4 =	2.48853E-02	1.69853E-02	7.74706E-03	2.35454E-02	-2.65798E-01
A6 =	4.08792E-02	-1.80144E-02	2.86908E-02	1.16717E-02	3.40106E-02
A8 =	-3.79010E-02	8.85295E-02	-7.51977E-02	9.37992E-03	-1.06362E-01
A10 =	4.11034E-02	-1.17837E-01	2.07031E-02	-6.69673E-02	3.44400E-02
A12 =	1.67658E-02	-6.44208E-02	-1.19308E-01	-2.93589E-02	-4.71228E-04
A14 =	-2.82099E-02	3.89946E-02	5.95103E-02	9.29812E-02	1.08030E-02
Surface #	7	8	9	10	11
k =	-1.00000E+00	-1.00000E+00	-7.74878E-01	-1.62472E+00	-1.00000E+00
A4 =	-2.30672E-01	-9.92656E-02	5.10610E-02	1.17135E-01	2.99395E-02
A6 =	8.97416E-02	4.67196E-02	-1.44127E-02	-4.48627E-02	-1.66520E-02
A8 =	-7.58480E-02	1.34449E-02	3.48387E-02	7.79041E-03	2.30890E-03
A10 =	5.60688E-02	-1.71060E-02	-1.17268E-02	-5.61845E-04	1.00768E-06
A12 =	-2.76063E-02	6.39548E-03	1.10625E-03	6.85778E-05	-2.74264E-05
A14 =	1.76742E-02	-1.09595E-03	-3.19965E-05	-4.58590E-06	-2.08896E-06
A16 =				-1.09314E-06	6.45068E-07

With reference to Table 19 and FIG. 7B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.20$ (mm), an overall aperture value (f-number) $Fno=2.40$, and a half of maximum view angle $HFOV=34.1^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 21 below, and the related symbols have been described above and thus will not be described again.

TABLE 21

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	1.12
$v_1 - v_2$	32.1
$v_1 - (v_2 + v_3)$	8.3
R_9/f	-0.25
$(R_7 + R_8)/(R_7 - R_8)$	1.01
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.20
T_{45}/f	0.14
f/f_3	-1.94
f/f_4	1.91
S_D/T_D	0.93
$TTL/ImgH$	1.73

According to the optical data as shown in Table 19 and the aberration curves as shown in FIG. 7B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

<Eighth Preferred Embodiment>

With reference to FIGS. 8A and 8B for a schematic view and a series of aberration curves of an optical image lens

assembly in accordance with the eighth preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, an aperture stop **800** and an IR-filter **860**. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: an aperture stop **800**; a plastic first lens element **810** with positive refractive power, having a convex object-side surface **811** and a concave image-side surface **812**, and both object-side surface **811** and image-side surface **812** thereof being aspheric; a plastic second lens element **820** with negative refractive power, having a convex object-side surface **821** and a concave image-side surface **822**, and both object-side surface **821** and image-side surface **822** thereof being aspheric; a plastic third lens element **830** with positive refractive power, having a concave object-side surface **831** and a convex image-side surface **832**, and both object-side surface **831** and image-side surface **832** being aspheric; a plastic fourth lens element **840** with positive refractive power, having a concave object-side surface **841** and a convex image-side surface **842**, and both object-side surface **841** and image-side surface **842** being aspheric; a plastic fifth lens element **850** with negative refractive power, having a concave object-side surface **851** and a convex image-side surface **852**, and both object-side surface **851** and image-side surface **852** thereof being aspheric, and at least one of the object-side surface **851** and image-side surface **852** thereof having at least one inflection point; a glass IR-filter **860**, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor **880** installed at an image plane **870**. With the combination of the five lens elements, the aperture stop **800** and the IR-filter **860**, an image of the object to be photographed can be formed at the image sensor **880**.

TABLE 24

Optical data of this preferred embodiment $f = 4.43$ mm, $Fno = 2.60$, $HFOV = 32.9$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.261				
2	Lens 1	1.480920 (ASP)	0.544	Plastic	1.544	55.9	2.77
3		81.272200 (ASP)	0.053				

TABLE 24-continued

Optical data of this preferred embodiment f = 4.43 mm, Fno = 2.60, HFOV = 32.9 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
4	Lens 2	5.048300 (ASP)	0.263	Plastic	1.650	21.4	-5.09
5		1.957960 (ASP)	0.477				
6	Lens 3	-6.167500 (ASP)	0.497	Plastic	1.544	55.9	16.64
7		-3.773200 (ASP)	0.355				
8	Lens 4	-4.116500 (ASP)	1.018	Plastic	1.544	55.9	2.47
9		-1.099870 (ASP)	0.347				
10	Lens 5	-0.961790 (ASP)	0.335	Plastic	1.544	55.9	-1.89
11		-17.211700 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.286				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

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The optical data of this preferred embodiment are listed in Table 24, wherein the object-side surface and the image-side surface of the first lens element **810** to the fifth lens element **850** comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 25 as follows:

TABLE 25

Aspheric coefficients of this preferred embodiment					
Surface #	2	3	4	5	6
k =	-4.79847E-01	0.00000E+00	-5.00000E+00	-1.28734E+00	-5.00000E+00
A4 =	2.68970E-02	4.28222E-02	7.93145E-03	2.31891E-02	-1.03610E-01
A6 =	2.12348E-02	-1.24274E-02	1.73523E-02	6.58890E-02	5.14651E-02
A8 =	-1.35494E-02	3.15467E-02	-2.53967E-02	2.38277E-02	-1.23470E-01
A10 =	4.29630E-02	-4.50252E-02	5.24752E-02	-4.81567E-03	8.45489E-02
A12 =	-3.08309E-02	1.79904E-02	-1.13257E-01	1.75175E-02	1.24324E-02
A14 =	1.47229E-02	-1.63474E-02	1.56180E-02	1.20738E-03	3.12046E-02
Surface #	7	8	9	10	11
k =	-3.68087E+00	1.00645E+01	-7.98743E-01	-1.58165E+00	-1.00000E+00
A4 =	-1.51522E-01	-1.51042E-01	6.84111E-02	1.21321E-01	1.66915E-02
A6 =	4.70183E-02	4.63751E-02	-1.71058E-02	-4.27485E-02	-1.50486E-02
A8 =	-8.12518E-02	-1.21211E-03	3.25058E-02	7.52027E-03	2.34732E-03
A10 =	5.53698E-02	-1.20053E-02	-1.23175E-02	-6.43800E-04	-1.79566E-05
A12 =	-3.26902E-02	1.10281E-02	1.07314E-03	5.80897E-05	-3.22185E-05
A14 =	1.64786E-02	-4.95565E-03	1.10610E-04	-4.05881E-06	-2.36412E-06
A16 =				-4.29490E-08	8.19756E-07

With reference to Table 24 and FIG. 8B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length f=4.43 (mm), an overall aperture value (f-number) Fno=2.60, and a half of maximum view angle HFOV=32.9°. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 26 below, and the related symbols have been described above and thus will not be described again.

TABLE 26

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	1.14
$v_1 - v_2$	34.5
$v_1 - (v_2 + v_3)$	-21.4

TABLE 26-continued

Data of related relations of this preferred embodiment	
Relation	Data
R_9/f	-0.22
$(R_7 + R_8)/(R_7 - R_8)$	1.73
$(R_9 + R_{10})/(R_9 - R_{10})$	-1.12
T_{45}/f	0.08
f/f_5	-2.34
f/f_4	1.79
S_D/T_D	0.93
$TTL/ImgH$	1.74

According to the optical data as shown in Table 24 and the aberration curves as shown in FIG. 8B, the optical imaging lens assembly in accordance with this preferred embodiment

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of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

<Ninth Preferred Embodiment>

With reference to FIGS. 9A and 9B for a schematic view and a series of aberration curves of an optical image lens assembly in accordance with the ninth preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, an aperture stop 900 and an IR-filter 960. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: an aperture stop 900; a plastic first lens element 910 with positive refractive power, having a convex object-side surface 911 and a convex image-side surface 912, and both object-side surface 911 and image-side surface 912 thereof being aspheric; a plastic second lens element 920 with negative refractive power, having a convex object-side surface 921 and a concave image-side surface 922, and both

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object-side surface 921 and image-side surface 922 thereof being aspheric; a plastic third lens element 930 with positive refractive power, having a concave object-side surface 931 and a convex image-side surface 932, and both object-side surface 931 and image-side surface 932 being aspheric; a plastic fourth lens element 940 with positive refractive power, having a concave object-side surface 941 and a convex image-side surface 942, and both object-side surface 941 and image-side surface 942 being aspheric; a plastic fifth lens element 950 with negative refractive power, having a concave object-side surface 951 and a concave image-side surface 952, and both object-side surface 951 and image-side surface 952 thereof being aspheric, and at least one of the object-side surface 951 and image-side surface 952 thereof having at least one inflection point; a glass IR-filter 960, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor 980 installed at an image plane 970. With the combination of the five lens elements, the aperture stop 900 and the IR-filter 960, an image of the object to be photographed can be formed at the image sensor 980.

TABLE 27

Optical data of this preferred embodiment f = 4.36 mm, Fno = 2.58, HFOV = 33.1 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.245				
2	Lens 1	1.537140 (ASP)	0.541	Plastic	1.544	55.9	2.78
3		-93.321600 (ASP)	0.083				
4	Lens 2	8.475600 (ASP)	0.258	Plastic	1.650	21.4	-5.22
5		2.392310 (ASP)	0.483				
6	Lens 3	-4.601700 (ASP)	0.732	Plastic	1.544	55.9	16.40
7		-3.206100 (ASP)	0.430				
8	Lens 4	-16.603400 (ASP)	0.788	Plastic	1.544	55.9	2.28
9		-1.173230 (ASP)	0.249				
10	Lens 5	-1.008210 (ASP)	0.335	Plastic	1.544	55.9	-1.65
11		9.345800 (ASP)	0.600				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.286				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

The optical data of this preferred embodiment are listed in Table 27, wherein the object-side surface and the image-side surface of the first lens element 910 to the fifth lens element 950 comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 28 as follows:

TABLE 28

Aspheric coefficients of this preferred embodiment					
Surface #	2	3	4	5	6
k =	-4.70073E-01	1.00000E+02	-5.00000E+00	-1.94574E+00	4.61178E-01
A4 =	2.75624E-02	3.74869E-02	-2.60359E-03	1.47472E-02	-1.05959E-01
A6 =	1.97564E-02	-1.64857E-02	8.22007E-03	4.64604E-02	4.09340E-02
A8 =	-8.31019E-03	3.42619E-02	-2.11831E-02	2.04546E-02	-1.30248E-01
A10 =	3.60011E-02	-4.74857E-02	6.05183E-02	3.73003E-03	8.65006E-02
A12 =	-3.08206E-02	1.79904E-02	-1.13257E-01	1.75175E-02	1.24392E-02
A14 =	1.47229E-02	-1.63474E-02	1.56180E-02	1.20738E-03	3.12046E-02
Surface #	7	8	9	10	11
k =	-1.69932E+00	7.30718E+01	-7.87674E-01	-1.65264E+00	-1.00000E+00
A4 =	-1.55872E-01	-1.94245E-01	6.59688E-02	1.19636E-01	-1.09722E-02
A6 =	5.25804E-02	2.90886E-02	-1.80552E-02	-4.26393E-02	-1.04576E-02

TABLE 28-continued

Aspheric coefficients of this preferred embodiment					
A8 =	-7.12227E-02	3.61147E-03	3.26318E-02	7.60087E-03	2.23618E-03
A10 =	6.02928E-02	-8.71349E-03	-1.23000E-02	-6.31923E-04	-6.93179E-05
A12 =	-3.65624E-02	1.17580E-02	1.06168E-03	5.83293E-05	-3.40002E-05
A14 =	1.31230E-02	-5.19556E-03	1.01428E-04	-4.21234E-06	-1.97031E-06
A16 =				-1.10947E-07	9.46402E-07

With reference to Table 27 and FIG. 9B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.36$ (mm), an overall aperture value (f-number) $Fno=2.58$, and a half of maximum view angle $HFOV=33.1^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 29 below, and the related symbols have been described above and thus will not be described again.

TABLE 29

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	1.16
$v_1 - v_2$	34.5
$v_1 - (v_2 + v_3)$	-21.4
R_9/f	-0.23
$(R_7 + R_8)/(R_7 - R_8)$	1.15
$(R_9 + R_{10})/(R_9 - R_{10})$	-0.81
T_{45}/f	0.06
f/f_5	-2.64
f/f_4	1.91
S_D/T_D	0.94
$TTL/ImgH$	1.75

According to the optical data as shown in Table 27 and the aberration curves as shown in FIG. 9B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

<Tenth Preferred Embodiment>

With reference to FIGS. 10A and 10B for a schematic view and a series of aberration curves of an optical image lens

assembly in accordance with the tenth preferred embodiment of the present invention respectively, the optical imaging lens assembly comprises five non-cemented lenses with refractive power, an aperture stop **1000** and an IR-filter **1060**. More specifically, the optical imaging lens assembly, sequentially arranged from an object side to an image side along an optical axis, comprises: a plastic first lens element **1010** with positive refractive power, having a convex object-side surface **1011** and a convex image-side surface **1012**, and both object-side surface **1011** and image-side surface **1012** thereof being aspheric; an aperture stop **1000**; a plastic second lens element **1020** with negative refractive power, having a concave object-side surface **1021** and a concave image-side surface **1022**, and both object-side surface **1021** and image-side surface **1022** thereof being aspheric; a plastic third lens element **1030** with negative refractive power, having a concave object-side surface **1031** and a convex image-side surface **1032**, and both object-side surface **1031** and image-side surface **1032** being aspheric; a plastic fourth lens element **1040** with positive refractive power, having a convex object-side surface **1041** and a convex image-side surface **1042**, and both object-side surface **1041** and image-side surface **1042** being aspheric; a plastic fifth lens element **1050** with negative refractive power, having a concave object-side surface **1051** and a concave image-side surface **1052**, and both object-side surface **1051** and image-side surface **1052** thereof being aspheric, and at least one of the object-side surface **1051** and image-side surface **1052** thereof having at least one inflection point; a glass IR-filter **1060**, made of panel glass for adjusting a wavelength section of the light of an image; and an image sensor **1080** installed at an image plane **1070**. With the combination of the five lens elements, the aperture stop **1000** and the IR-filter **1060**, an image of the object to be photographed can be formed at the image sensor **1080**.

TABLE 30

Optical data of this preferred embodiment $f = 4.52$ mm, $Fno = 2.43$, $HFOV = 32.0$ deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Lens 1	1.665870 (ASP)	0.695	Plastic	1.544	55.9	2.67
2		-9.631400 (ASP)	0.017				
3	Ape. Stop	Plano	0.111				
4	Lens 2	-15.955800 (ASP)	0.260	Plastic	1.633	23.4	-4.37
5		3.363200 (ASP)	0.501				
6	Lens 3	-5.185800 (ASP)	0.373	Plastic	1.633	23.4	-31.38
7		-7.214400 (ASP)	0.611				
8	Lens 4	34.356000 (ASP)	0.821	Plastic	1.544	55.9	1.92
9		-1.067820 (ASP)	0.249				
10	Lens 5	-1.154740 (ASP)	0.350	Plastic	1.535	56.3	-1.71
11		4.830900 (ASP)	0.700				
12	IR-filter	Plano	0.300	Glass	1.517	64.2	
13		Plano	0.278				
14	Image	Plano	—				

Note:

Reference wavelength is 587.6 nm.

ASP stands for aspherical surfaces.

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The optical data of this preferred embodiment are listed in Table 30, wherein the object-side surface and the image-side surface of the first lens element **1010** to the fifth lens element **1050** comply with the aspheric surface formula as given in Equation (14), and their aspheric coefficients are listed in Table 31 as follows:

TABLE 31

Aspheric coefficients of this preferred embodiment					
Surface #	1	2	4	5	6
k =	-7.05176E-01	-1.00000E+00	2.64902E+02	-1.00000E+00	-5.57502E+01
A4 =	1.77464E-02	4.95414E-02	4.97176E-02	-1.28393E-02	-2.71828E-01
A6 =	2.29635E-02	-3.95821E-02	-6.26360E-03	1.60346E-02	8.57194E-02
A8 =	-4.47880E-02	4.01476E-02	-3.21105E-03	3.02444E-02	-1.22238E-01
A10 =	5.63473E-02	-3.82602E-02	7.98554E-02	-6.66269E-02	4.79552E-02
A12 =	-3.40211E-02	1.11513E-02	-1.31953E-01	4.80269E-02	3.72205E-02
A14 =	6.31528E-03	2.86369E-03	8.41915E-02	7.66567E-09	7.64028E-09
Surface #	7	8	9	10	11
k =	-1.00000E+00	2.00000E+02	-1.03629E+00	-2.79849E+00	-1.00000E+00
A4 =	-1.97287E-01	-1.44461E-01	1.02012E-01	9.55319E-02	-2.53796E-02
A6 =	1.02856E-01	5.63103E-02	-4.14257E-02	-4.22517E-02	-6.81526E-03
A8 =	-8.92083E-02	1.05429E-02	3.49400E-02	7.74576E-03	1.61792E-03
A10 =	6.54822E-02	-1.70016E-02	-1.19017E-02	-6.54463E-04	-6.48162E-05
A12 =	-1.81216E-02	6.53248E-03	1.05322E-03	5.55113E-05	-2.87615E-05
A14 =	7.88890E-03	-1.32392E-03	4.18746E-05	-3.58259E-06	-1.56491E-06
A16 =				-1.15024E-07	7.98686E-07

With reference to Table 30 and FIG. 10B for an optical imaging lens assembly of this preferred embodiment, the optical imaging lens assembly has a focal length $f=4.52$ (mm), an overall aperture value (f-number) $Fno=2.43$, and a half of maximum view angle $HFOV=32.0^\circ$. After the optical data of this preferred embodiment are calculated and derived, the optical imaging lens assembly satisfies related conditions as shown in Table 32 below, and the related symbols have been described above and thus will not be described again.

TABLE 32

Data of related relations of this preferred embodiment	
Relation	Data
R_8/R_9	0.92
$v_1 - v_2$	32.5
$v_1 - (v_2 + v_3)$	9.1
R_9/f	-0.26
$(R_7 + R_8)/(R_7 - R_8)$	0.94
$(R_9 + R_{10})/(R_9 - R_{10})$	-0.61
T_{45}/f	0.06
f/f_5	-2.64
f/f_4	2.35
S_D/T_D	0.82
$TTL/lmgH$	1.81

According to the optical data as shown in Table 32 and the aberration curves as shown in FIG. 10B, the optical imaging lens assembly in accordance with this preferred embodiment of the present invention provides good correction results in aspects of the longitudinal spherical aberration, astigmatic field curving, and distortion.

In the optical imaging lens assembly of the present invention, the lens element can be made of glass or plastic. If the lens element is made of glass, the selection of refractive power for the optical imaging lens assembly can be more flexible. If the lens element is made of plastic, then the production cost can be reduced effectively.

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In the optical imaging lens assembly of the present invention, if the lens element has a convex surface, then the surface of the lens element is convex at a position near the optical axis; and if the lens element has a concave surface, then the surface of the lens element is concave at a position near the optical axis.

In the optical imaging lens assembly of the present invention, at least one stop such as a glare stop or a field stop can be provided for reducing stray lights to improve the image quality.

Tables 1 to 32 show changes of values of an optical imaging lens assembly in accordance with different preferred embodiments of the present invention respectively, and even if different values are used, products of the same structure are intended to be covered by the scope of the present invention. It is noteworthy to point out that the aforementioned description and the illustration of related drawings are provided for the purpose of explaining the technical characteristics of the present invention, but not intended for limiting the scope of the present invention.

What is claimed is:

1. An optical imaging lens assembly, comprising five non-cemented lenses with refractive power, sequentially arranged from an object side to an image side comprising:

a first lens element with positive refractive power;

a second lens element with negative refractive power;

a third lens element with refractive power having a concave object-side surface, and at least one of the object-side surface and an image-side surface thereof being aspheric;

a plastic fourth lens element with positive refractive power having both aspheric object-side surface and image-side surface; and

a plastic fifth lens element with negative refractive power having a concave object-side surface and a convex image-side surface, and both object-side surface and image-side surface thereof being aspheric;

wherein f is a focal length of the optical imaging lens assembly, f_4 is a focal length of the fourth lens element, R_7 is a curvature radius of the object-side surface of the fourth lens element, R_8 is a curvature radius of the image-side surface of the fourth lens element, and the following relations are satisfied:

$$1.55 < f/f_4 < 3.50; \text{ and}$$

$$0.4 < (R_7 + R_8)/(R_7 - R_8) < 2.5.$$

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2. The optical imaging lens assembly of claim 1, wherein the first lens element has a convex object-side surface, and the fourth lens element has a convex image-side surface.

3. The optical imaging lens assembly of claim 2, wherein R_9 is a curvature radius near the optical axis of the object-side surface of the fifth lens element, f is the focal length of the optical imaging lens assembly, and the following relation is satisfied:

$$-0.5 < R_9/f < 0.$$

4. The optical imaging lens assembly of claim 3, further comprising a stop, wherein T_D is an axial distance from the object-side surface of the first lens element to the image-side surface of the fifth lens element, S_D is an axial distance from the stop to the image-side surface of the fifth lens element, and the following relation is satisfied:

$$0.70 < S_D/T_D < 1.20.$$

5. The optical imaging lens assembly of claim 4, wherein f is the focal length of the optical imaging lens assembly, f_5 is a focal length of the fifth lens element, and the following relation is satisfied:

$$-3.0 < f/f_5 < -1.7.$$

6. The optical imaging lens assembly of claim 4, wherein v_1 is an Abbe number of the first lens element, v_2 is an Abbe number of the second lens element, and the following relation is satisfied:

$$28 < v_1 - v_2 < 45.$$

7. The optical imaging lens assembly of claim 3, wherein T_{45} is an axial distance between the fourth lens element and the fifth lens element, f is the focal length of the optical imaging lens assembly, and the following relation is satisfied:

$$0.02 < T_{45}/f < 0.18.$$

8. The optical imaging lens assembly of claim 3, wherein v_1 is an Abbe number of the first lens element, v_2 is an Abbe number of the second lens element, v_3 is an Abbe number of the third lens element, and the following relation is satisfied:

$$-10 < v_1 - (v_2 + v_3) < 20.$$

9. The optical imaging lens assembly of claim 3, wherein R_9 is the curvature radius of the object-side surface of the fifth lens element, f is the focal length of the optical imaging lens assembly, and the following relation is satisfied:

$$-0.3 < R_9/f < 0.$$

10. The optical imaging lens assembly of claim 3, wherein R_9 is the curvature radius of the object-side surface of the fifth lens element, R_{10} is a curvature radius of the image-side surface of the fifth lens element, and the following relation is satisfied:

$$-2.0 < (R_9 + R_{10})/(R_9 - R_{10}) < -1.02.$$

11. The optical imaging lens assembly of claim 3, further comprising an image sensor at an image plane for imaging an object to be photographed, wherein TTL is a distance from the object-side surface of the first lens element to the image plane, $ImgH$ is half of the diagonal length of an effective photosensitive area of the image sensor, and the following relation is satisfied:

$$TTL/ImgH < 1.95.$$

12. The optical imaging lens assembly of claim 3, wherein at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point.

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13. The optical imaging lens assembly of claim 2, wherein R_8 is the curvature radius of the image-side surface of the fourth lens element, R_9 is a curvature radius of the object-side surface of fifth lens element, and the following relation is satisfied:

$$0.9 < R_8/R_9 < 1.25.$$

14. The optical imaging lens assembly of claim 2, wherein R_7 is the curvature radius of the object-side surface of the fourth lens element, R_8 is the curvature radius of the image-side surface of the fourth lens element, and the following relation is satisfied:

$$0.6 < (R_7 + R_8)/(R_7 - R_8) < 1.9.$$

15. An optical imaging lens assembly, comprising five non-cemented lenses with refractive power, sequentially arranged from an object side to an image side comprising:

- a first lens element with positive refractive power;
- a second lens element with negative refractive power;
- a third lens element with negative refractive power having at least one aspheric object-side surface and image-side surface;
- a plastic fourth lens element with positive refractive power having both aspheric object-side surface and image-side surface; and
- a plastic fifth lens element with negative refractive power having a concave object-side surface and a convex image-side surface, and both object-side surface and image-side surface thereof being aspheric;

wherein R_7 is a curvature radius of the object-side surface of the fourth lens element, R_8 is a curvature radius of the image-side surface of the fourth lens element, and the following relation is satisfied:

$$0.4 < (R_7 + R_8)/(R_7 - R_8) < 2.5.$$

16. The optical imaging lens assembly of claim 15, wherein R_9 is a curvature radius of the object-side surface of fifth lens element, f is a focal length of the optical imaging lens assembly, and the following relation is satisfied:

$$-0.5 < R_9/f < 0.$$

17. The optical imaging lens assembly of claim 16, wherein f is the focal length of the optical imaging lens assembly, f_4 is a focal length of the fourth lens element, f_5 is a focal length of the fifth lens element, and the following relations are satisfied:

$$1.55 < f/f_4 < 3.50; \text{ and}$$

$$-3.0 < f/f_5 < -1.7.$$

18. The optical imaging lens assembly of claim 16, wherein v_1 is an Abbe number of the first lens element, v_2 is an Abbe number of the second lens element, v_3 is an Abbe number of the third lens element, and the following relation is satisfied:

$$-10 < v_1 - (v_2 + v_3) < 20.$$

19. The optical imaging lens assembly of claim 16, wherein R_8 is the curvature radius of the image-side surface of the fourth lens element, R_9 is the curvature radius of the object-side surface of fifth lens element, and the following relation is satisfied:

$$0.9 < R_8/R_9 < 1.25.$$

20. The optical imaging lens assembly of claim 16, wherein R_9 is the curvature radius of the object-side surface of the fifth lens element, R_{10} is a curvature radius of the image-side surface of the fifth lens element, and the following relation is satisfied:

$$-2.0 < (R_9 + R_{10})/(R_9 - R_{10}) < -1.02.$$

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21. The optical imaging lens assembly of claim 16, wherein at least one of the object-side surface and the image-side surface of the fifth lens element has at least one inflection point.

22. An optical imaging lens assembly, comprising five non-cemented lenses with refractive power, sequentially arranged from an object side to an image side, comprising:

- a first lens element with positive refractive power;
- a second lens element with negative refractive power;
- a third lens element with refractive power having at least one aspheric surface;
- a plastic fourth lens element with positive refractive power having both aspheric object-side surface and image-side surface; and
- a plastic fifth lens element with negative refractive power having a concave object-side surface and a convex image-side surface, and both object-side surface and image-side surface thereof being aspheric, and at least one of the object-side surface and image-side surface thereof having at least one inflection point;

wherein f is a focal length of the optical imaging lens assembly, f_4 is a focal length of the fourth lens element, R_9 is a curvature radius of the object-side surface of fifth lens element, and the following relations are satisfied:

$1.55 < f/f_4 < 3.50$; and

$-0.5 < R_9/f < 0$.

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23. The optical imaging lens assembly of claim 22, wherein R_7 is a curvature radius of the object-side surface of the fourth lens element, R_8 is a curvature radius of the image-side surface of the fourth lens element, and the following relation is satisfied:

$0.6 < (R_7 + R_8)/(R_7 - R_8) < 1.9$.

24. The optical imaging lens assembly of claim 22, wherein R_8 is a curvature radius of the image-side surface of the fourth lens element, R_9 is the curvature radius of the object-side surface of fifth lens element, and the following relation is satisfied:

$0.9 < R_8/R_9 < 1.25$.

25. The optical imaging lens assembly of claim 22, wherein f is the focal length of the optical imaging lens assembly, f_4 is the focal length of the fourth lens element, f_5 is a focal length of the fifth lens element, and the following relations are satisfied:

$1.55 < f/f_4 < 3.50$; and

$-3.0 < f/f_5 < -1.7$.

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